SUPERCONDUCTING QUANTUM HARDWARE ENGINEERING

A Practical Design Monograph for Microwave Engineers: From Planar Chip Layout to Cryogenic Packaging, RF Instrumentation, and 3D Scientific Visualization

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Specification Parameter	Project Definition
Author / Operating Framework	Antigravity Engineering Meta-Framework (SYZYGY Operating Loop)
Technical Classification	Applied Microwave Engineering • Superconducting cQED • Cryogenic Systems
Design Methodology	Deterministic Spec-Driven Hardware Synthesis (Zero Quantum Physics Fluff)
Target CAD & Software Stack	Qiskit Metal • KLayout • Ansys HFSS / AWS Palace • KiCad • Blender 4.x
Edition & Format	Monochrome Laboratory Reference Edition (Pure Academic Black & White)
Curriculum Target	Complete Hardware Replication Manual for Šimon Marek Quantum Series

EXECUTIVE ARCHITECTURE ABSTRACT

MONOGRAPH ABSTRACT: This engineering manual provides an exhaustive, practical roadmap for replicating high-performance superconducting quantum processors, microwave packaging enclosures, and photorealistic 3D scientific animations. Traditional quantum computing literature relies heavily on abstract quantum mechanics formalism (Hamiltonian perturbation expansions, Hilbert space state vectors, and master equations in the Lindblad form), which obscures the tangible engineering parameters needed to design physical chips. In this manual, the superconducting transmon qubit is treated strictly as an *RF non-linear circuit component* governed by distributed transmission line impedance, capacitive matrices, nonlinear Josephson inductance, S-parameters, and electromagnetic cavity boundary conditions.

Across six comprehensive chapters and technical appendices, this document details all five phases of development: (1) Lumped specifications and the four golden design numbers ($f_{01}, \alpha, f_r, \chi$); (2) Planar chip CAD, electron beam lithography, Dolan bridge junctions, 3D electromagnetic finite-element simulation, and pyEPR automated quantization; (3) Microwave sample packaging, OFHC copper cavity puck sizing, Rogers 4350B PCB interposers, and wirebonding interconnects; (4) Dilution refrigerator thermal attenuation budgeting (-60 dB), JTWPA parametric amplifiers, discrete 4–8 GHz cryogenic LNA PCB design, and FPGA-based direct digital synthesis using Xilinx RFSoC; and (5) The complete Blender 3D scientific visualization pipeline, including physically based rendering (PBR) materials, three-point studio lighting, camera kinematics, and exploded-view product rigging.

All calculations are presented with explicit numerical values, mechanical tolerances, cleanroom recipes, and direct Python scripts.

TABLE OF CONTENTS & CURRICULUM ARCHITECTURE

Section	Title & Module Scope	Technical Deliverables & Reference Standards
Study Guide	Curriculum Syllabus & Pre-Watch Video Matrix	Video 1 to 9 preparation: target numbers, mental models, what to ignore
Chapter 1	Phase 1: Specifications & Lumped-Element Model	Lumped models, Transmon physics, cQED foundations, Noise budget
Chapter 2	Phase 2: Planar Chip CAD & 3D EM Simulation	Qiskit Metal, KLayout, Dolan lithography, HFSS/Palace, pyEPR quantization
Chapter 3	Phase 3: Microwave Packaging & Carrier PCB	Cavity box modes, OFHC copper pucks, Rogers 4350B CPWG, Wirebonding
Chapter 4	Phase 4: Cryogenic Hardware & RF Electronics	Dilution fridges, 60dB attenuation, JTWPA, 4–8 GHz LNA PCB, RFSoC control
Chapter 5	Phase 5: 3D Scientific Visualization in Blender	CAD import, PBR shaders (Si, Nb, Au), Studio lighting, Exploded rigging
Chapter 6	Phase 6: 2D Surface Code Multi-Qubit Architecture	Coupling buses, Cross-Resonance gates, 17-qubit lattice, Airbridges, TSVs
Chapter 7	Phase 7: Nanofabrication Metrology & Packaging SOP	Ambegaokar-Baratoff wafer probing, SEM metrology, Fastener torque specs
Chapter 8	Phase 8: Master Numerical Reference & Diagnostics	20-parameter golden design, Troubleshooting matrix, Sign-off checklist
Appendix A	Cleanroom Lithography Run-Sheet & Recipes	Spin coating, bake profiles, EBL doses, oxidation, liftoff chemistry
Appendix B	Golden Formula & Unit Conversion Cheat Sheet	Microwave formulas, conversion constants (GHz to μeV, dBm to photons/s)
Appendix C	Glossary of Quantum Hardware Terminology	cQED, box modes, DRAG, JTWPA, Purcell, OFHC, SQUID, TSV definitions

SYSTEM IMPLEMENTATION GUIDELINES

HOW TO NAVIGATE THIS MANUAL:

• **Video Companion Mode:** Read the Pre-Watch Syllabus (Pages 3–6) before watching each YouTube module to lock in target numbers.

• **For Chip Designers:** Study Chapters 1 and 2. Follow the numerical walkthrough to calculate pad area and junction critical current, then execute the provided Python scripts to export GDSII polygons and automate pyEPR in HFSS.

• **For Microwave PCB & Packaging Engineers:** Study Chapter 3. Apply the cavity box mode cutoff formula ($f_{110} > 15\ \mathrm{GHz}$) to ensure the metallic sample puck does not induce parasitic Purcell relaxation, and apply the CPWG via stitching rules in KiCad.

• **For Cryogenic & RF Instrumentation Engineers:** Study Chapter 4. Implement the -60 dB coaxial attenuator tree inside the dilution refrigerator and design the 4–8 GHz LNA PCB matching network using Rogers RO4350B substrate.

• **For Scientific Animators & Visualizers:** Study Chapter 5. Import the chip vector layout directly into Blender, configure the PBR material shaders for silicon, niobium thin film, and gold-plated copper, and set up the camera and exploded-view keyframes.

CURRICULUM SYLLABUS

Video-by-Video Pre-Watch Study Roadmap

What to Study in this Manual Before Hitting Play on Each Module of the Series

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HOW TO USE THIS MANUAL WITH ŠIMON MAREK'S VIDEOS

HOW TO USE THIS MANUAL WITH THE VIDEO SERIES:

This laboratory manual is designed as an interactive companion to Šimon Marek's 9-part quantum hardware and 3D scientific visualization series. To maximize technical comprehension and avoid theoretical rabbit holes, **do not watch the videos blindly**. Before hitting play on each video, review the corresponding chapter, lock in the exact target numbers, and understand what CAD clicks or equations are about to appear on screen.

The roadmap below provides the precise reading assignment, key parameters, mental models, and non-essential theory to ignore for each video in the series.

Master Pre-Watch Study & Target Parameters Matrix

Video Module	Module Title	Exact Manual Reading Assignment	Key Numerical Targets & Concepts
Video 1	Transmon Qubit — How Do They Work?	Chapter 1 (Pages 4–7) + Appendix B	f01 = 5.0 GHz, α = -250 MHz, EJ/EC ≥ 50, LJ0 = 11.86 nH, χ = 227 kHz
Video 2	Designing a Quantum Amplifier (JTWPA) in HFSS	Chapter 4.3 (Pages 21–22) + Chapter 2.4	Four-Wave Mixing (2ωp = ωs + ωi), +20 dB gain, quantum noise limit (■ω/2)
Video 3	Superconducting Qubit Layout (cQED)	Chapter 2.1–2.3 (Pages 8–10)	50 Ω CPW (w = 10 μm, s = 6 μm), λ/4 meander (l = 4.13 mm), claw Cc = 5.8 fF
Video 4	Simulating Qubits with pyEPR & HFSS	Chapter 2.4 (Pages 10–12)	Linear eigenmode substitution (LJ0), Energy Participation Ratio (pj), α extraction
Video 5	Microwave Packaging & Carrier Board	Chapter 3 (Pages 13–17) + Chapter 8.2	Cavity cutoff fbox > 15 GHz, Indium seal, Rogers 4350B CPWG, Lwire < 0.8 nH
Video 6	4–8 GHz Cryogenic LNA Design in KiCad	Chapter 4.4 (Pages 22–23)	Discrete pHEMT at 4 K (Tn ≈ 2 K), gate/drain bias-tees, stability K > 1, SMT passives
Video 7	FPGA Quantum Control (RFSoC / QICK)	Chapter 4.5 (Pages 23–25)	9.85 GS/s RF-DAC direct synthesis, 20 ns DRAG envelope, 4.0 GS/s ADC DDC, < 80 ns latency
Video 8	Surface Code Multi-Qubit Chip & Holder	Chapter 7 (Pages 31–33)	λ/2 coupling buses (geff), Cross-Resonance CNOT, airbridges, TSVs, 17-qubit rotated lattice
Video 9	3D Scientific Illustration in Blender	Chapter 5 (Pages 26–30)	Silicon PBR (IOR 3.9), 3-point studio lighting, f = 65–85 mm DoF f/2.8, 180-frame storyboard

Detailed Video-by-Video Study Protocols (Modules 1 to 4)

MODULE 1: TRANSMON QUBIT PHYSICS & LUMPED MODEL

Video 1: Transmon Qubit — How Do They Work?

• **What the video shows:** Wavefunction probability distributions, Cooper pair tunneling across an insulating barrier, the cosine Josephson potential energy well, energy level quantization, and the Bloch sphere.

• **What to cover from PDF first:** Read **Chapter 1 (Pages 4–7)** and **Appendix B**. Study why a linear LC circuit climbs energy levels and why the non-linear Josephson junction creates the crucial negative inductance shift (anharmonicity α = -250 MHz). Review Section 1.4 on the EJ/EC ≥ 50 flat-band rule to see how shunt capacitance cancels charge noise.

• **What to IGNORE:** Do not get bogged down in wave mechanics, density matrices, Hilbert spaces, or Mathieu function derivations. Treat the transmon strictly as a non-linear microwave LC resonator.

MODULE 2: JTWPA PARAMETRIC AMPLIFIER IN HFSS

Video 2: Designing a Quantum Amplifier (JTWPA) using Ansys HFSS

• **What the video shows:** Setting up an electromagnetic 3D model in Ansys HFSS containing ~2000 cascaded Josephson junctions or high-kinetic-inductance nanowires to amplify single-photon microwave readout tones.

• **What to cover from PDF first:** Read **Section 4.3**. Understand why commercial 4 K HEMT amplifiers add 10 to 20 noise photons, which swamps weak readout signals (~ -125 dBm). Learn the Four-Wave Mixing condition: $2\omega p = \omega s + \omega i$, providing +20 dB of broadband gain operating at the fundamental quantum noise floor (■$\omega/2$).

• **What to IGNORE:** Non-linear quantum Langevin differential equations and vacuum fluctuation operators. Treat the JTWPA as a distributed parametric microwave pre-amplifier.

MODULE 3: CHIP CAD & cQED PLANAR LAYOUT

Video 3: Designing a Superconducting Quantum Processor (CAD & cQED Layout)

• **What the video shows:** Synthesizing polygons in Qiskit Metal or KLayout: transmon capacitor pads, Dolan junction pockets, capacitive coupling claws, and distributed coplanar waveguide meander lines.

• **What to cover from PDF first:** Read **Section 2.1–2.3**. Verify the substrate parameters (High-Resistivity Float-Zone Silicon, thickness 500 μm, εr = 11.7) and the 50 Ω CPW dimensions (trace width w = 10 μm, gap s = 6 μm). Review why the readout resonator is a quarter-wave ($\lambda/4$) meander line of length l = 4.13 mm.

• **Hands-On Step:** Run python transmon_calculator.py in your workspace to verify that pad capacitance $C\Sigma$ = 77.5 fF produces f01 = 5.000 GHz and α = -250 MHz.

MODULE 4: pyEPR QUANTIZATION IN HFSS

Video 4: Simulating Superconducting Qubits with pyEPR & Ansys HFSS

• **What the video shows:** Assigning material boundaries in HFSS, setting up an Eigenmode solution, and running the open-source pyEPR Python library to extract qubit frequency, anharmonicity, and dispersive shift.

• **What to cover from PDF first:** Read **Section 2.4**. Understand the *linear replacement technique*: the non-linear Josephson junction is replaced by a linear lumped inductor (LJ0 = 11.86 nH). HFSS solves the linear EM fields, and pyEPR computes the Energy Participation Ratio pj, directly extracting the full quantum Hamiltonian without non-linear time-domain differential equations.

• **What to IGNORE:** Multi-mode Hamiltonian perturbation expansions. Follow the step-by-step pyEPR workflow in Chapter 2.

Detailed Video-by-Video Study Protocols (Modules 5 to 9)

MODULE 5: MICROWAVE PACKAGING & CARRIER PCB

Video 5: Microwave Packaging & Carrier Board Design

• **What the video shows:** CNC-machining an OFHC copper puck cavity, designing a high-frequency Rogers PCB interposer, compressing an Indium wire gasket, and ultrasonic wedge wirebonding.

• **What to cover from PDF first:** Read **Chapter 3** and **Section 8.2**. Memorize the *Cavity Box Mode Cutoff Rule*: puck internal cavity pocket dimensions must be ≤ 8 mm × 8 mm × 3 mm so the lowest cavity resonance f110 = 26.5 GHz is safely > 15 GHz above the operating band. Review the Rogers RO4350B CPWG trace geometry and the Wirebond Inductance Rule (Lwire < 0.8 nH; loop length < 0.8 mm).

• **What to IGNORE:** Detailed cryogenic thermodynamic derivations. Focus on microwave box modes and impedance continuity.

MODULE 6: 4–8 GHz CRYOGENIC LNA DESIGN IN KICAD

Video 6: 4–8 GHz Cryogenic LNA Design in KiCad

• **What the video shows:** Schematics and PCB layout in KiCad: selecting discrete pHEMT transistors, gate/drain bias-tees, stability networks, and coplanar waveguide routing with via stitching.

• **What to cover from PDF first:** Read **Section 4.4**. Study the cryogenic LNA schematic and BOM table: Broadcom ATF-36077 pHEMT, gate bias Vgs = -0.22 V, drain bias Vds = +1.0 V at 10 mA, 10 Ω series gate resistor for unconditional Rollett stability (K > 1), and Rogers RO4350B substrate routing.

• **What to IGNORE:** Semiconductor solid-state physics and cryogenic carrier freeze-out mathematics. Focus on RF matching and biasing.

MODULE 7: FPGA RFSOC DIRECT MICROWAVE SYNTHESIS

Video 7: FPGA Quantum Control & Instrumentation (RFSoC / QICK)

• **What the video shows:** Connecting the dilution refrigerator to a Xilinx Zynq UltraScale+ RFSoC board, generating DRAG pulses, digitizing readout return tones, and discriminating |0■ vs |1■ IQ constellation clusters.

• **What to cover from PDF first:** Read **Section 4.5**. Understand the 3-stage DSP pipeline: (1) 14-bit 9.85 GS/s RF-DAC direct microwave synthesis in the 2nd Nyquist zone; (2) Real-time DRAG pulse envelope shaping (canceling state |2■ leakage); and (3) 4.0 GS/s RF-ADC Digital Downconversion (DDC) with matched filtering in FPGA DSP slices within 80 nanoseconds.

MODULE 8: SURFACE CODE MULTI-QUBIT LATTICE

Video 8: Surface Code Multi-Qubit Architecture & Lattice Holder

• **What the video shows:** A multi-qubit chip holder containing a 2D planar array of transmons, coupling bus meander lines, aluminum airbridge crossovers, and through-silicon vias (TSVs).

• **What to cover from PDF first:** Read **Chapter 7**. Study: (1) Distributed λ/2 CPW coupling buses mediating virtual photon exchange (geff ≈ 6.4 MHz); (2) All-microwave Cross-Resonance (CR) entangling CNOT gates with echo cancellation; and (3) 3D Superconducting Aluminum Airbridges (height 3.5 μm, span 35 μm) spaced every 120 μm to suppress parasitic slotline odd modes.

MODULE 9: BLENDER 3D SCIENTIFIC VISUALIZATION

Video 9: Scientific Illustration & 3D Hardware Animation in Blender

• **What the video shows:** Importing DXF/SVG and STEP CAD files into Blender 4.x, assigning PBR shaders, setting up 3-point studio lighting, configuring depth of field, and keyframing an exploded-view product animation.

• **What to cover from PDF first:** Read **Chapter 5**. Follow: (1) The PBR Material Table (Silicon Die: Base #101418, Roughness 0.04, **IOR = 3.90**); (2) 3-Point Studio Lighting (Key 50W, Fill 15W, **Rim 80W strip behind chip**); (3) Camera setup (f = 65–85 mm, Depth of Field f/2.8 focused on junction); and (4) The 180-frame exploded keyframe timeline table.

CHAPTER 1

The Non-Linear Circuit & Phase 1 Specifications

Lumped Element Models, Transmon Qubit Physics, and Circuit QED Foundations

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1.1 The Microwave Baseline: Why Linear LC Tanks Cannot Form a Qubit

Microwave engineers are intimately familiar with standard resonant tank circuits. A parallel or series LC resonator consists of a linear inductor L and a capacitor C, yielding a fundamental resonant frequency $\omega_0 = 1 / \sqrt{LC}$. When such an oscillator is cooled to temperatures below 20 mK—where thermal noise energy $k_B T$ is far smaller than the single-photon energy quantum $\hbar \omega_0$—its energy states quantize into discrete rungs: $E_n = \hbar \omega_0 (n + 1/2)$, where $n \in \{0, 1, 2, \dots\}$.

The critical limitation of this linear network is that the energy difference between any two adjacent states is strictly constant: $\Delta E = E_{n+1} - E_n = \hbar \omega_0$. If an external microwave source drives the circuit at frequency ω_0 to excite the system from the ground state $|0\rangle$ to the first excited state $|1\rangle$, the identical drive tone will simultaneously stimulate transitions from $|1\rangle \to |2\rangle$, $|2\rangle \to |3\rangle$, and upward through the entire infinite harmonic ladder. This phenomenon is termed **harmonic ladder climbing**. Because all energy transitions are spectrally degenerate, it is mathematically impossible to isolate two discrete computational states using a linear inductor. To build an addressable quantum bit, the microwave circuit must incorporate an **anharmonic (non-linear) inductor**.

1.2 The Josephson Junction: Physical Anatomy & Operating Principles

The only electronic component capable of providing non-dissipative non-linearity at cryogenic temperatures is the **Josephson Junction (JJ)**. Fabricated using shadow evaporation lithography (the Dolan bridge technique), a Josephson junction consists of two superconducting aluminum thin films separated by an ultra-thin amorphous aluminum oxide ($\mathrm{Al}_2\mathrm{O}_3$) barrier having a thickness of only 1 to 2 nanometers. Below aluminum's critical temperature ($T_c \approx 1.2\ \mathrm{K}$), conduction electrons bind into **Cooper pairs** ($2e$) that tunnel coherently across the insulating barrier without electrical resistance.

The behavior of the junction is completely defined by two fundamental equations linking current, voltage, and the macroscopic superconducting phase difference ϕ:

THE CONSTITUTIVE JOSEPHSON RELATIONS

1. First Josephson Relation (Current-Phase):

$$I(t) = I_c \cdot \sin(\phi(t))$$

2. Second Josephson Relation (Voltage-Phase):

$$V(t) = (\hbar / 2e) \cdot (d\phi / dt) = (\Phi_0 / 2\pi) \cdot (d\phi / dt)$$

where I_c is the junction critical current (typically 20 to 35 nA in transmon designs), and $\Phi_0 = h / (2e) \approx 2.0678 \times 10^{-15}$ *Wb* is the magnetic flux quantum.

To understand how this junction behaves as a non-linear inductor, we differentiate the first relation with respect to time: $dI/dt = I_c \cos(\phi) \cdot (d\phi/dt)$. Rearranging the second relation yields $d\phi/dt = (2\pi / \Phi_0) V(t)$. Substituting this gives:

Non-Linear Inductance Formula: $$L_J(\phi) = V / (dI / dt) = \Phi_0 / (2\pi I_c \cos(\phi)) = L_{J0} / \cos(\phi)$$

At zero bias current ($\phi = 0$), the small-signal zero-bias inductance is $L_{J0} = \Phi_0 / (2\pi I_c)$. For an engineered critical current $I_c = 30$ *nA*, $L_{J0} \approx 10.97$ *nH*. As current flows through the junction, the effective inductance increases as $1/\cos\phi$. This non-linear inductance shifts higher energy levels downward in frequency, breaking the harmonic degeneracy and creating the required **anharmonicity (α)**.

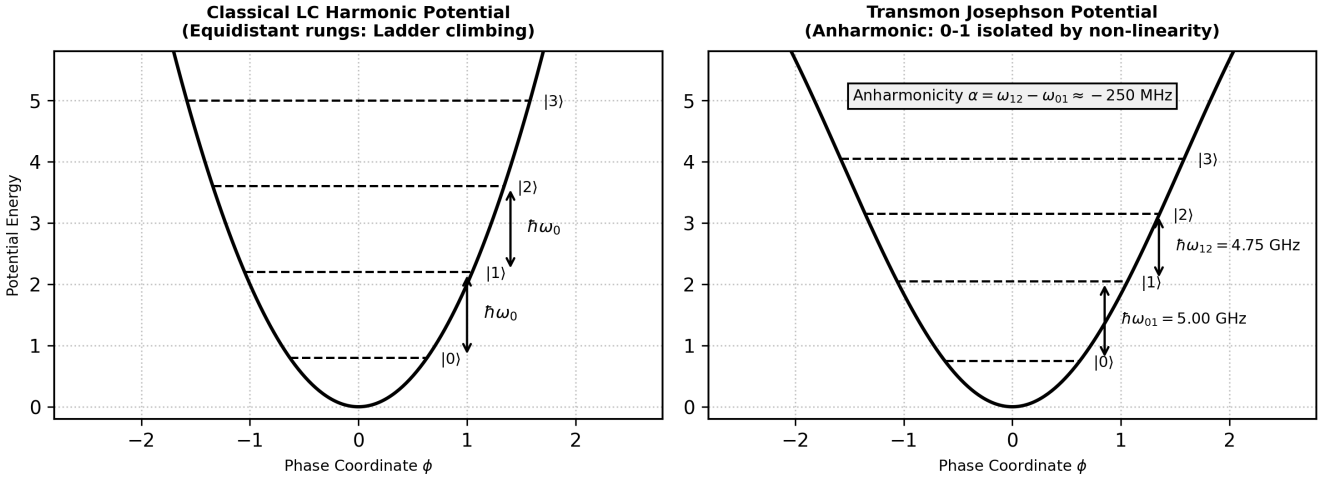


Figure: *Potential energy profile of a classical linear LC oscillator (left) showing equal energy level spacing, compared with the cosine potential of a transmon qubit (right) where non-linearity isolates the 0-1 transition by anharmonicity $\alpha \approx$ -250 MHz.*

1.3 The Transmon Architecture: Silencing Electrostatic Charge Noise

The earliest superconducting charge qubit was the *Cooper Pair Box (CPB)*, which paired a small Josephson junction with a tiny capacitance. While the CPB exhibited huge anharmonicity (several gigahertz), its resonant frequency was violently sensitive to ambient electrostatic charge fluctuations (n_g). Stray electrons hopping on the substrate or microphonic vibrations shifted the qubit frequency by hundreds of megahertz, causing rapid dephasing ($T_2^* < 1\ \mu\mathrm{s}$).

In 2007, Koch et al. at Yale University introduced the **Transmon** (Transmission line shunted plasma oscillation qubit). By placing two large planar capacitor pads in parallel with the junction, the total shunt capacitance is engineered to $C_\Sigma = C_J + C_B \approx 70\text{--}90\ \mathrm{fF}$. This reduces the single-electron charging energy $E_C = e^2 / (2 C_\Sigma)$ to approximately $200\text{--}300\ \mathrm{MHz}$. Simultaneously, the junction is sized such that the ratio of Josephson tunneling energy E_J to charging energy E_C satisfies the **Transmon Condition**:

The Golden Transmon Ratio: $E_J / E_C \geq 50$ **(Optimal operating envelope: 50 to 80)**

In this regime, the sensitivity to charge noise (the *charge dispersion* ϵ) is suppressed exponentially according to: $\varepsilon \propto exp(-\sqrt{(8\ E_J / E_C)})$. Operating at $E_J / E_C = 65$ suppresses charge dispersion from 2 GHz down to less than 1 kHz! The qubit's energy levels become completely flat with respect to gate charge, silencing dephasing and extending coherence times from nanoseconds to well over 100 microseconds.

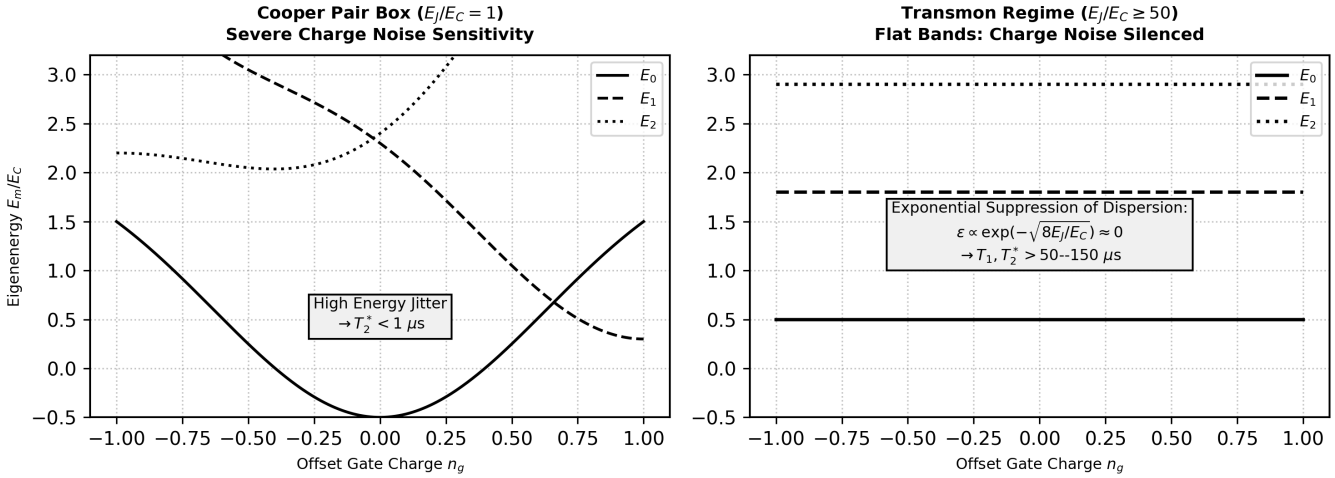


Figure: *Eigenenergies versus offset gate charge ng. In the Cooper Pair Box (left, EJ/EC = 1), energy levels fluctuate wildly with charge noise. In the Transmon regime (right, EJ/EC ≥ 50), the energy bands are completely flat, rendering the qubit immune to charge noise.*

1.4 The Duffing Non-Linearity & Fundamental Formulas

Expanding the cosine potential energy $U(\phi) = -E_J \cos\phi$ in a Taylor series about $\phi = 0$ yields: $U(\phi) \approx -E_J (1 - \phi^2/2 + \phi^4/24 - \dots)$. The quadratic term $\phi^2/2$ corresponds to a harmonic oscillator with plasma frequency $\omega_p = \sqrt{8 E_J E_C} / \hbar$. The negative quartic term $-\phi^4/24$ is identical to the **Duffing non-linearity** (or 3rd-order intermodulation distortion, IP_3) in microwave amplifiers. This term produces the **negative anharmonicity (α)**:

Parameter	Engineering Equation	Physical Implementation Context
Charging Energy (E_C)	$E_C = e^2 / (2 C_\Sigma)$	Single-electron electrostatic charging energy (~200–300 MHz)
Josephson Energy (E_J)	$E_J = (\Phi_0 I_c) / (2\pi)$	Cooper pair tunneling energy (~12–18 GHz)
Qubit Transition (f_{01})	$f_{01} \approx (\sqrt{(8 E_J E_C)} - E_C) / h$	Frequency needed to drive qubit from |0■ to |1■ (~4.8–5.2 GHz)
Anharmonicity (α)	$\alpha = f_{12} - f_{01} \approx -E_C / h$	Frequency separation between 0-1 and 1-2 transitions (~-200 to -300 MHz)

Zero-Bias Inductance (L_{J0})	$L_{J0} = \Phi_0 / (2\pi I_c)$	Equivalent linear lumped inductance of junction (~8–14 nH)

1.5 Circuit QED (cQED) & Dispersive Readout

Directly attaching a coaxial cable to a transmon qubit collapses its quantum state within picoseconds due to environmental loading. To measure the qubit non-destructively, we implement **Circuit Quantum Electrodynamics (cQED)**: the qubit is capacitively coupled to a high-Q distributed Coplanar Waveguide (CPW) transmission line resonator.

By choosing the detuning between qubit frequency ω_q and resonator frequency ω_r to be much larger than their coupling rate g ($|\Delta| = |\omega_q - \omega_r| \gg g$), real photon exchange is forbidden. Instead, the interaction enters the **dispersive regime**. The presence of the qubit pulls the resonant frequency of the readout resonator by a dispersive shift $\pm\chi$:

Dispersive Shift Formula: $\chi \approx (g^2 \cdot E_C) / (\Delta \cdot (\Delta - E_C))$

where g is the capacitive coupling rate (typically 50 to 80 MHz), $\Delta = \omega_q - \omega_r$ is the detuning (typically -1.5 to -2.2 GHz), and E_C is the charging energy (250 MHz). This produces a typical dispersive shift $\chi \approx 0.2\text{--}1.0\ \mathrm{MHz}$.

When an RF probe tone is launched at the bare resonator frequency $\omega_{\mathrm{probe}} = \omega_r$, the transmitted transmission coefficient (S_{21}) experiences a state-dependent phase shift. Demodulating this microwave signal using an IQ mixer yields two distinct voltage clusters on the complex plane, allowing deterministic single-shot qubit state determination.

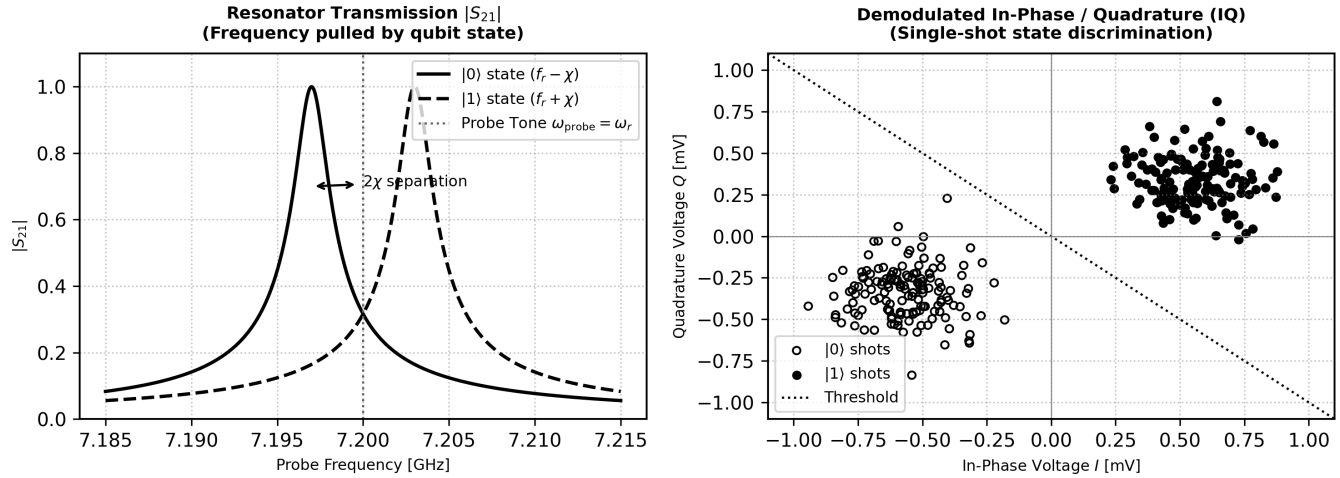


Figure: *Dispersive readout spectroscopy. (Left) Resonator transmission magnitude |S21| shifts by 2χ between qubit states |0■ and |1■. (Right) Demodulated In-phase / Quadrature (IQ) plane showing single-shot separation between states.*

1.6 Coherence, Noise Channels & The Quality Factor Budget

A quantum chip's performance is quantified by its coherence time constants: **T_1 (energy relaxation time)**, representing how long state $|1\rangle$ persists before decaying to $|0\rangle$, and **T_2^* (dephasing time)**, representing phase stability. In microwave engineering terms, T_1 corresponds directly to the loaded quality factor of the non-linear resonator: $Q = \omega_{01} T_1$. A qubit with $T_1 = 100\ \mu\mathrm{s}$ at 5.0 GHz possesses an extraordinary loaded quality factor $Q = 2\pi \times (5 \times 10^9) \times (100 \times 10^{-6}) \approx 3.14 \times 10^6$.

To achieve multi-million quality factors, the engineer must budget against four primary loss channels:

Loss Channel	Physical Mechanism	Loss Formulation	Engineering Mitigation in CAD / Hardware
Purcell Radiation Loss	Energy leaking through readout resonator into 50Ω feedline	$T_{1,Purcell} \approx (\Delta / g)^2 \cdot (Q_{ext} / \omega_r)$	Maintain $\|\Delta\| \geq 2.0$ GHz, $Q_{ext} \geq 10{,}000$; implement bandpass Purcell filter

Two-Level System (TLS) Loss	Dielectric defect dipoles at metal-substrate and surface oxides	$1/T_{1,TLS} = p_{interface} \cdot \omega_{01} \cdot tan\ \delta$	Make capacitor pad gaps large (>35μm) to dilute surface electric field density
Quasiparticle Poisoning	Broken Cooper pairs caused by stray infrared radiation or cosmic rays	$1/T_{1,qp} \propto n_{qp} \cdot \sqrt{(2\Delta_{gap} / \blacksquare\omega)}$	Enclose sample in multi-layer light-tight copper pucks lined with ECCOSORB absorber
Magnetic Flux Noise	Low-frequency (1/f) magnetic flux fluctuations modulating SQUID loop	$1/T_{\phi} \propto \lvert\partial\omega_{01} / \partial\Phi\rvert \cdot A_{\Phi}$	Use fixed-frequency transmons without SQUIDs, or shield with Cryoperm / Mu-metal cylinders

1.7 Master Design Table: The 4 Target Numbers & Physical Knobs

Every quantum chip design starts by fixing four fundamental specifications. The table below links these numbers directly to physical CAD dimensions:

Design Metric	Nominal Value	Tolerance Window	Physical Hardware Knob in Layout
Qubit Frequency (f_{01})	5.000 GHz	4.80 – 5.20 GHz	Junction area (I_c) and capacitor pad surface area (C_Σ)
Anharmonicity (α)	-250.0 MHz	-200 to -300 MHz	Total pad capacitance C_Σ; $\alpha = -e^2 / (2\ h\ C_\Sigma)$
Readout Frequency (f_r)	7.200 GHz	6.80 – 7.50 GHz	Physical length of λ/4 CPW meander line: $l = c_0 / (4\ f_r\ \sqrt{(\varepsilon_{eff})})$
Dispersive Shift (χ)	227 kHz	150 – 500 kHz	Claw coupling capacitance C_g setting mutual coupling rate g

CHAPTER 2

Planar Chip CAD & 3D Electromagnetic Simulation

Qiskit Metal, KLayout, Dolan Lithography, Ansys HFSS, Palace, and pyEPR Quantization

2.1 Substrate & Metallization Engineering

Superconducting circuits operate at gigahertz frequencies where bulk dielectric loss dominates. Standard FR4 ($\tan\delta \sim 0.02$) or alumina substrates are entirely unacceptable. Instead, quantum chips are fabricated on **High-Resistivity Float-Zone Silicon (HR-Si)** having resistivity $\rho > 10\ \mathrm{k}\Omega\cdot\mathrm{cm}$, relative permittivity $\varepsilon_r = 11.7$, and intrinsic dielectric loss tangent $\tan\delta < 10^{-6}$, or on c-plane Sapphire ($\mathrm{Al}_2\mathrm{O}_3$, $\varepsilon_r = 10.0$). Standard wafer thickness is $h = 500\ \mu\mathrm{m}$.

For the metallization layer, two superconducting thin films dominate: **1. Niobium (Nb):** Sputtered thin film ($150\text{--}200\ \mathrm{nm}$). Critical temperature $T_c = 9.2\ \mathrm{K}$. Extremely rugged, resistant to oxidation, and ideal for base wiring, coplanar waveguides, and readout resonators. **2. Aluminum (Al):** Electron-beam evaporated thin film ($30\text{--}100\ \mathrm{nm}$). Critical temperature $T_c = 1.2\ \mathrm{K}$. Mandatory for Josephson junctions because aluminum forms a self-limiting, atomically uniform amorphous oxide barrier ($\mathrm{Al}_2\mathrm{O}_3$).

2.2 Coplanar Waveguide (CPW) Transmission Line Engineering

All microwave signals on chip propagate via Coplanar Waveguide (CPW) transmission lines consisting of a center conductor of width w separated by gaps of width s from two semi-infinite ground planes. For an HR-Si substrate ($\varepsilon_r = 11.7$), the effective dielectric constant for a thick substrate is given by: $\varepsilon_{eff} = (\varepsilon_r + 1) / 2 = (11.7 + 1) / 2 = 6.35$.

The standard 50 Ω transmission line geometry utilizes: **Center width w = 10 μm, gap s = 6 μm**. The phase velocity of microwave propagation is $v_p = c_0 / \sqrt{\varepsilon_{\mathrm{eff}}} = (3 \times 10^8) / \sqrt{6.35} \approx 1.19 \times 10^8\ \mathrm{m/s}$. A quarter-wavelength ($\lambda/4$) readout resonator operating at $f_r = 7.200\ \mathrm{GHz}$ has physical length:

Resonator Length Calculation: $l = v_p / (4 \cdot f_r) = (1.19 \times 10^8) / (4 \times 7.20 \times 10^9) = 4.13\ \mathrm{mm}\ (4130\ \mu\mathrm{m})$

To fit a 4.13 mm resonator onto a small silicon chip, the CPW trace is meandered. **Routing Design Rules:** (1) Maintain bend radius $R_{\mathrm{bend}} \ge 3w = 30\ \mu\mathrm{m}$ to prevent current crowding and radiation loss; (2) Keep spacing between adjacent meander arms $\ge 5s = 30\ \mu\mathrm{m}$ to prevent parasitic coupling that shifts the resonant frequency.

2.3 Transmon Pocket & Capacitor Pad Geometry

The transmon pocket consists of an etched ground plane opening containing two isolated rectangular metal pads connected by the Josephson junction. The total capacitance C_Σ is the sum of pad-to-pad mutual capacitance C_{mutual} and pad-to-ground capacitance C_{ground}. To achieve $C_\Sigma = 77.5\ \mathrm{fF}$ on silicon: • Pocket cutout opening: $300\ \mu\mathrm{m} \times 450\ \mu\mathrm{m}$; • Twin capacitor pads: $240\ \mu\mathrm{m} \times 180\ \mu\mathrm{m}$ each, separated by a $40\ \mu\mathrm{m}$ central gap; • Readout coupling claw: A CPW finger wrapped around the top pad with a $15\ \mu\mathrm{m}$ separation gap, yielding mutual capacitance $C_g \approx 3.5\ \mathrm{fF}$ ($g \approx 70\ \mathrm{MHz}$).

2.4 Automated Programmatic Layout with Qiskit Metal (Python)

Instead of drawing polygons manually in AutoCAD, production quantum chips are generated parametrically in Python using **Qiskit Metal**. The complete script below builds the entire planar chip geometry:

PYTHON LAYOUT GENERATION CODE (QISKIT METAL)

```
# Qiskit Metal Parametric Layout Script (Transmon + Readout CPW)
from qiskit_metal import designs, MetalGUI
from qiskit_metal.qlibrary.qubits.transmon_pocket import TransmonPocket
from qiskit_metal.qlibrary.tlines.meandered import ResonatorMeander
from qiskit_metal.qlibrary.tlines.straight_path import RouteStraight

design = designs.DesignPlanar()
design.chips.main.size.size_x = '6mm'
design.chips.main.size.size_y = '6mm'

# 1. Instantiate Transmon Pocket with twin capacitor pads
q1 = TransmonPocket(design, 'Q1', options=dict(
    pos_x='-1.5mm', pos_y='0.0mm',
    pad_width='240um', pad_height='180um', pad_gap='40um',
    inductor_width='10um', junction_area='0.03um2',
    pocket_width='300um', pocket_height='450um'))

# 2. Instantiate λ/4 CPW Readout Resonator (Target fr = 7.2 GHz)
r1 = ResonatorMeander(design, 'R1', options=dict(
    total_length='4.13mm', trace_width='10um', trace_gap='6um',
    meander=dict(spacing='100um', asymmetry='0um')))

# 3. Export GDSII layout for e-beam lithography
gds = design.renderers.gds
gds.export_to_gds('transmon_chip_v1.gds')
print('GDSII successfully exported for cleanroom fabrication.')
```

2.5 Detailed Cleanroom Fabrication Recipe: Dolan Bridge Process

Josephson junctions cannot be etched with standard optical lithography because their critical dimensions ($150\text{ nm} \times 200\text{ nm}$) fall below the diffraction limit of light. They are fabricated using **Electron-Beam Lithography (EBL)** and the **Dolan double-angle shadow evaporation process**. The complete cleanroom standard operating procedure (SOP) is detailed in the run-card below:

Fabrication Step	Cleanroom Tool & Process Parameters	Engineering Function
1. Wafer Clean	Piranha etch (3:1 H2SO4:H2O2) at 120°C for 10 min; 10:1 BOE dip for 30s	Removes organic contaminants and native silicon oxide (SiO2) layer
2. Base Metallization	DC magnetron sputtering of 150 nm Niobium at 3 µbar Argon pressure	Zero-stress superconducting base layer for CPW lines and ground plane
3. CPW Patterning	Spin AZ1518 positive photoresist (1.5 µm); UV contact exposure (120 mJ/cm²)	Patterns readout feedlines, resonators, and transmon ground cutouts
4. Reactive Ion Etch	SF6 / Ar plasma RIE (25 sccm SF6, 10 sccm Ar, 100 W RF power, 20 mTorr)	Anisotropic dry etch of Niobium down to silicon surface (rate ~80 nm/min)
5. EBL Bilayer Resist	Spin MMA(8.5)EL10 at 4000 RPM (bake 180°C, 5 min); spin PMMA 950K at 4000 RPM	Bottom MMA layer has high EBL sensitivity; top PMMA layer forms rigid bridge
6. E-Beam Exposure	100 kV Raith / JEOL EBL tool. Undercut dose: 450 µC/cm²; Bridge dose: 250 µC/cm²	Creates undercut resist cavity with suspended PMMA Dolan bridge

7. Development	Develop in MIBK:IPA (1:3) at 20°C for 60 seconds; rinse in pure IPA for 30 seconds	Cleans out exposed resist cavity; inspect bridge integrity under optical microscope
8. In-Situ Ar Cleaning	Argon ion mill in evaporation chamber (30 sec, 200 V beam voltage, 10 mA current)	Strips native surface oxides from base niobium contact pads prior to aluminum deposition
9. Layer 1 Aluminum	E-beam evaporation of 30 nm pure Aluminum at -45° tilt angle (rate 1.0 nm/s)	Forms bottom electrode of the Josephson junction in UHV (< 10■■ mbar)
10. Thermal Oxidation	Static oxidation: leak pure O2 at 10 mbar pressure for 12 minutes at 22°C	Grows controlled 1.5 nm amorphous Al2O3 tunnel barrier (sets Ic ≈ 28 nA)
11. Layer 2 Aluminum	E-beam evaporation of 60 nm pure Aluminum at +45° tilt angle (rate 1.0 nm/s)	Forms overlapping top electrode of the Josephson junction
12. Liftoff & Strip	Soak in N-methyl-2-pyrrolidone (NMP) at 80°C for 2 hours; gentle sonication	Removes remaining resist bilayer, leaving completed sub-micron junctions

2.6 3D Electromagnetic Simulation with Ansys HFSS / AWS Palace

Before committing designs to cleanroom fabrication, the 3D model must be simulated in a finite-element electromagnetic solver. The complete step-by-step Standard Operating Procedure (SOP) is detailed below:

ANSYS HFSS 3D SIMULATION SOP

2.8 Step-by-Step 3D Simulation SOP in Ansys HFSS:
To simulate a planar transmon chip die inside Ansys HFSS from scratch, execute the following protocol:
Step 1: Project & Coordinate Setup: Open Ansys Electronics Desktop. Insert an HFSS design. Set model units to micrometers (μm). Establish the global coordinate system at the chip substrate center.
Step 2: Substrate Geometry: Create a 3D box for the substrate: Dimension X = 6000 μm, Y = 6000 μm, Z = -500 μm. Assign material as High-Resistivity Silicon (relative permittivity εr = 11.7, bulk dielectric loss tangent tan δ = 1.0 × 10■■).
Step 3: Ground Plane & Thin-Film Traces: Draw 2D rectangular sheets on the top substrate surface (Z = 0). Draw the ground plane opening (pocket: 300 μm × 450 μm). Draw the twin capacitor pads (240 μm × 180 μm separated by 40 μm). Draw the readout CPW meander trace (width w = 10 μm, gap s = 6 μm). Assign all 2D metal sheets as Perfect Electric Conductor (PEC) boundary conditions.
Step 4: Non-Linear Junction Lumped Substitution: In the 40 μm central gap between capacitor pads, draw a 2D rectangle (width = 10 μm, length = 40 μm). Assign this rectangle as a **Lumped RLC Boundary**: Set R = 0 Ω, C = 0 F, and L = LJ0 = 11.86 nH. Define the integration line arrow pointing directly from Pad 1 to Pad 2.
Step 5: Vacuum Enclosure & Radiation Boundaries: Create an air/vacuum box surrounding the chip: Dimension X = 8000 μm, Y = 8000 μm, Z = 3000 μm (extending 2000 μm above the chip and 1000 μm below). Assign the outer faces of the vacuum box as Radiation Boundaries (or PEC boundaries if simulating enclosed metallic box modes).
Step 6: Solution Setup (Eigenmode Analysis): Add an Eigenmode Solution Setup. Set Minimum Frequency = 3.0 GHz, Number of Modes = 3. Mode 1 will solve for the Transmon mode (target ~5.0 GHz), Mode 2 will solve for the Readout Resonator mode (target ~7.2 GHz), and Mode 3 will capture parasitic chip/cavity modes. Set Maximum Delta Frequency Per Pass = 0.1% to guarantee mesh convergence.
Step 7: Adaptive Meshing Refinement: Add a Length-Based Mesh Operation to the transmon capacitor pads and CPW slots. Restrict maximum element length to < 4.0 μm near the junction gap. This ensures electric field gradients across the junction are resolved.
Step 8: Execution & Field Export: Solve the project (typically 8 to 15 adaptive passes). Once converged, inspect the electric field magnitude |E| overlay on the substrate. Launch pyEPR to parse the solution and extract multi-mode Hamiltonian parameters.

Simulation Setting	Nominal Value / Setup Rule	Physical Verification Purpose
Mesh Element Size	< 4.0 μm near junction; < 20 μm on ground planes	Ensures high-gradient fringing fields near transmon pads are resolved
Convergence Criterion	Maximum Delta Frequency Δf < 0.1% per pass	Guarantees numerical stability of extracted capacitance and eigenmode
Boundary Conditions	PEC for 2D metal; Lumped RLC (LJ0 = 11.86 nH) for junction	Models superconducting metal lossless conduction and Josephson inductance
Substrate Definition	Silicon (εr = 11.7, tan δ = 1.0 × 10■■, thickness 500 μm)	Accounts for substrate dielectric loading and microwave phase velocity
Vacuum Radiation Box	Air box extending 2 mm above and 1 mm below substrate	Absorbs open radiated energy without spurious metallic wall reflections

2.7 Automated Hamiltonian Quantization via pyEPR

Classical EM solvers only compute linear electromagnetic fields. They cannot natively calculate non-linear quantum transitions. The **Energy Participation Ratio (pyEPR)** method (Minev et al., Yale/IBM) solves this: HFSS computes linear eigenmode field profiles $\vec{E}_m, \vec{H}_m$. pyEPR calculates what fraction of the total electromagnetic energy of mode m is stored inside the junction inductance: $p_{mj} = E_{inductive,\ junction} / E_{total,\ mode}$. From this linear ratio, pyEPR analytically extracts the full non-linear quantum Hamiltonian: qubit frequency f_{01}, anharmonicity α, coupling g, and dispersive shift χ in seconds without simulating quantum wavefunctions.

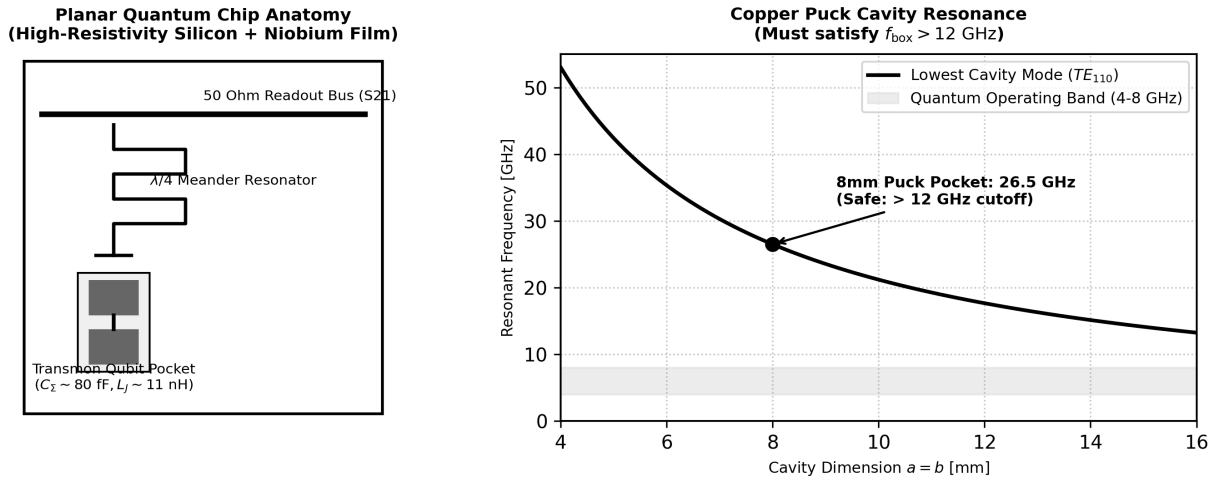


Figure: *Planar chip layout topology on silicon (left) and resonant cutoff frequency of the copper sample holder cavity versus box size (right). Keeping puck cavities under 8mm ensures box modes resonate safely above 25 GHz.*

CHAPTER 3

Microwave Packaging & Carrier PCB Interposer

Cavity Box Modes, OFHC Copper Pucks, Rogers 4350B CPWG Interposers, and Wirebonding

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3.1 Physics of Chip Enclosures & Cavity Box Modes

A quantum processor cannot operate in open space. It must be mechanically anchored and shielded from stray electromagnetic radiation. However, enclosing a dielectric chip inside a metallic enclosure creates a **3D microwave resonant cavity**. For a rectangular metallic box of internal dimensions $a \times b \times c$ (width, length, height), the lowest resonant mode (TE_{110}) is:

Cavity Box Mode Cutoff Equation: $$f_{110} = (c_0 / 2) \cdot \sqrt{((1 / a^2) + (1 / b^2))}$$

The Critical Failure Mechanism: If cavity dimensions are large (e.g., $a = b = 20\ \mathrm{mm}$), the box resonates at $f_{110} = 10.6\ \mathrm{GHz}$, and higher-order modes drop directly into the $5\text{--}7\ \mathrm{GHz}$ qubit operating band! If any box mode aligns with a qubit frequency, the qubit radiates its energy into the metallic walls, collapsing coherence ($T_1 < 100\ \mathrm{ns}$).

The Packaging Rule of Thumb: Size the internal chip pocket such that $f_{box} > 15$ to 25 GHz. For an $8\text{ mm} \times 8\text{ mm} \times 3\text{ mm}$ cavity pocket, $f_{110} = 26.50\ \mathrm{GHz}$. This pushes all cavity box resonances far above the quantum operating spectrum, completely preventing cavity Purcell decay.

3.2 Oxygen-Free High-Conductivity (OFHC) Copper Sample Puck

The physical puck is CNC-machined from **Oxygen-Free High-Conductivity Copper (OFHC, Alloy C10100)** with >99.99% pure copper content. OFHC copper maintains exceptional thermal conductivity at milli-Kelvin temperatures ($k > 1000\ \mathrm{W/(m\cdot K)}$ at 4 K). All surfaces are electroplated with $2\text{--}3\ \mu\mathrm{m}$ of high-purity gold over an electroless nickel barrier to prevent surface oxidation and ensure low contact resistance.

Seam Loss Suppression: Currents crossing the interface between the puck base and lid dissipate power if contact resistance is non-zero. To eliminate seam resistance: (1) Use perimeter clamping screws spaced every 5 mm; (2) Compress a $0.5\ \mathrm{mm}$ diameter pure Indium wire gasket in a milled groove along the joint seam. Indium becomes superconducting below $3.4\ \mathrm{K}$, creating a lossless continuous electrical seal.

3.3 High-Frequency Carrier PCB Design in KiCad / Altium

The silicon die is mounted inside a center cutout of a carrier PCB interposer that routes signals from coaxial connectors to the chip edges. Standard FR4 cannot be used due to severe dielectric losses at milli-Kelvin temperatures. The interposer is fabricated on **Rogers RO4350B or RO4003C** hydrocarbon ceramic laminate ($\varepsilon_r = 3.55$, $\tan\delta = 0.002$, thickness $h = 0.508\ \mathrm{mm}$).

Interposer Parameter	Design Specification	Microwave & Cryogenic Justification
Transmission Topology	Coplanar Waveguide with Ground (CPWG)	Center conductor on top, coplanar ground planes, solid ground plane on bottom layer
50 Ω Trace Geometry	Width W = 0.43 mm, Gap S = 0.25 mm	Achieves precise 50.0 Ω characteristic impedance on 0.508 mm Rogers 4350B substrate
Ground Via Fencing	Via pitch ≤ 1.5 mm (λ/10 at 10 GHz)	Continuous ground vias connecting top and bottom ground planes; suppresses slotline modes
Coaxial Connectors	SMP / SMPM blind-mate or edge-launch SMA	Rated up to 18–40 GHz; low insertion loss (<0.1 dB), return loss S11 < -20 dB across band

Plating Finish	ENIG or ENEPIG (Immersion Gold)	Electroless Nickel / Immersion Gold (ENIG) ensures reliable aluminum wirebonding

3.4 Step-by-Step Carrier PCB Design SOP in KiCad

To layout the Rogers 4350B interposer board in KiCad with controlled microwave impedance and wirebonding pads, follow this protocol:

KICAD CARRIER PCB DESIGN PROTOCOL

3.5 Step-by-Step Carrier PCB Design SOP in KiCad:
To design the high-frequency Rogers RO4350B PCB interposer for the quantum puck in KiCad, execute the following protocol:
Step 1: Board Stackup Configuration: Open KiCad Board Setup. Configure a 4-layer controlled-impedance stackup: • Layer 1 (F.Cu): 1 oz Copper (35 μm), ENIG gold plated (top microwave signal and coplanar ground planes); • Dielectric 1: Rogers RO4350B core (thickness h = 0.508 mm, εr = 3.66, tan δ = 0.0037); • Layer 2 (In1.Cu): Solid Copper Ground Plane (35 μm); • Dielectric 2: FR4 prepreg (thickness h = 0.600 mm) for mechanical rigidity; • Layer 3 (In2.Cu): DC bias routing and auxiliary low-frequency lines; • Layer 4 (B.Cu): Bottom solid ground shielding plane.
Step 2: 50 Ω CPWG Trace Routing: Use KiCad's built-in PCB Calculator tool to synthesize Coplanar Waveguide with Ground (CPWG): For substrate h = 0.508 mm, εr = 3.66, set center trace width W = 0.43 mm, gap to ground S = 0.25 mm. Route all microwave lines with smooth 45° mitered corners or curved circular arcs (bend radius R ≥ 1.5 mm).
Step 3: Ground Via Stitching Fences: Place continuous rows of grounded plated through-hole (PTH) vias (drill diameter = 0.3 mm, pad diameter = 0.6 mm) along both sides of every CPWG trace. Maintain via-to-via pitch P ≤ 1.5 mm (less than λ/10 at 10 GHz). This creates an effective coaxial shielding channel that prevents slotline mode excitation and suppresses crosstalk to < -50 dB.
Step 4: Chip Cavity Pocket & Edge Connector Footprints: Create a rectangular milling cutout in the board center (dimensions 6.2 mm × 6.2 mm) where the silicon chip die will sit directly on the copper puck base. Terminate the PCB traces at the edge of the cutout with rectangular wirebonding pads (0.3 mm × 0.6 mm) plated with ENIG gold. At the outer perimeter, place footprints for edge-mount high-frequency SMA connectors (Rosenberger 32K243-40ML5) or vertical blind-mate SMP/SMPM connectors (Amphenol / Carlisle). Verify S11 < -20 dB across 4 to 8 GHz.

3.5 Wirebonding & Interconnect Engineering

Connections between the Rogers PCB interposer and the silicon chip die are bridged using an ultrasonic wedge wirebonder with **99.99% pure Aluminum wire ($25\ \mu\mathrm{m}$ diameter)**. Aluminum is chosen because it becomes superconducting below $1.2\ \mathrm{K}$, eliminating resistive loss at the connection interface.

The Wirebond Inductance Rule: Each millimeter of wirebond adds approximately $1\ \mathrm{nH}$ of parasitic series inductance ($L_{\mathrm{wire}} \approx 1\ \mathrm{nH/mm}$). At 7 GHz, a 2 mm wirebond presents a reactive impedance $X_L = 2\pi f L = 2\pi \times (7 \times 10^9) \times (2 \times 10^{-9}) \approx 88\ \Omega$, causing severe reflection ($S_{11} > -6\ \mathrm{dB}$) and phase distortion! **Design Rule:** Keep wirebond loop lengths strictly under ***0.8 mm (L_{wire} < 0.8 nH)***. Furthermore, stitch 4 to 6 ground return wirebonds on both sides of every signal launch to maintain continuous coplanar ground return paths.

Interconnect Parameter	Design Constraint	Electromagnetic & Reliability Justification
Bondwire Material	99.99% Pure Aluminum (AlSi 1%)	Superconducting below 1.2 K; zero ohmic loss at milli-Kelvin
Wire Diameter	25 μm (0.001 inch)	Standard wedge bonding tooling diameter; high mechanical compliance
Maximum Loop Height	< 150 μm above chip surface	Minimizes vertical inductive loop area and external magnetic pickup
Maximum Span Length	< 800 μm (0.8 mm)	Ensures series inductance Lwire < 0.8 nH (S11 < -20 dB across 4–8 GHz)
Ground Return Ratio	4 to 6 ground wires per signal bond	Maintains ground return continuity; suppresses slotline ground bounce

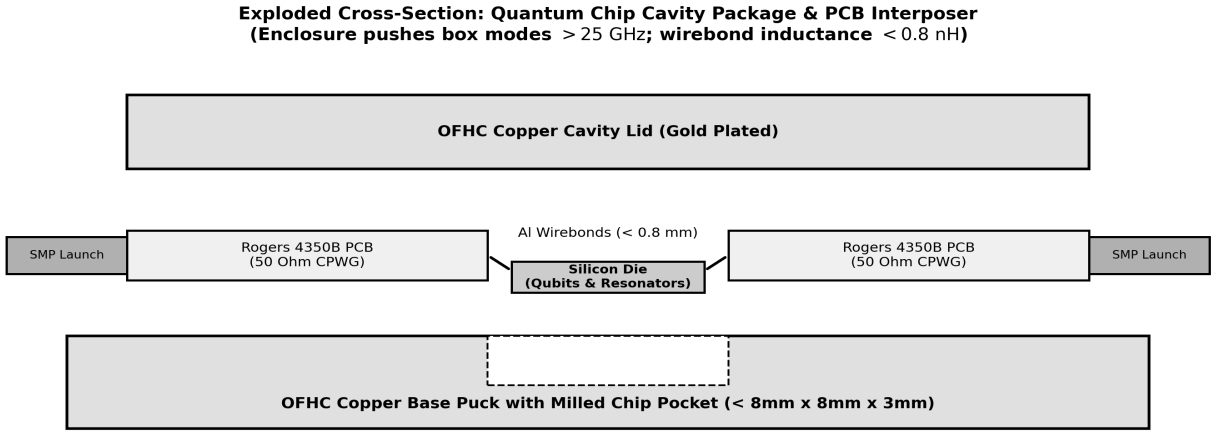


Figure: *Exploded cross-section of the quantum processor packaging assembly. The OFHC copper base puck, Rogers 4350B interposer PCB, silicon chip die, aluminum wirebond array, and indium-sealed cavity lid ensure box modes remain above 25 GHz.*

CHAPTER 4

Cryogenic Hardware & Discrete RF Electronics

Dilution Refrigerators, Attenuation Trees, JTWPA Amplifiers, 4-8 GHz LNA PCB, and RFSoC

4.1 The Dilution Refrigerator Environment: Thermal Stages & Heat Budgets

Superconducting transmons require ambient thermal energy to be far below the qubit energy splitting: $k_B T \ll \hbar \omega_{01}$. At 5.0 GHz, $\hbar \omega_{01} / k_B \approx 240\ \mathrm{mK}$. Operating at $15\ \mathrm{mK}$ ensures thermal excitation probability is negligible ($P_{|1\rangle} \approx \exp(-\hbar\omega / k_B T) = \exp(-240 / 15) \approx 1.1 \times 10^{-7}$). This temperature is achieved using a **$^3\mathrm{He}/^4\mathrm{He}$ Dilution Refrigerator (DR)** (e.g., Bluefors XLD or Oxford Instruments Proteox).

The cryostat is structured into five distinct thermal stages, each with strict cooling power budgets:

Thermal Flange	Primary Function	Cooling Power Budget	Microwave Component Allocation
50 K Stage (Radiation Shield)	300 K radiation interception	~40 W cooling power	Thermal anchor for outer stainless-steel coaxial lines and DC wiring
4 K Stage (Pulse Tube Flange)	Primary heat dissipation stage	~1.5 W cooling power	Houses cryogenic HEMT Low-Noise Amplifiers (dissipating ~10 mW each)
Still Stage (800 mK)	3He evaporation distillation	~20 mW cooling power	Intermediate thermalization point for coaxial lines and attenuators
Cold Plate (100 mK)	Heat-exchanger condensation	~1 mW cooling power	Hosts cryogenic microwave circulators and isolators (20 dB isolation)
Mixing Chamber (MXC, 15 mK)	Phase separation quantum base	15 – 30 μW cooling power	Houses OFHC copper puck with quantum processor inside magnetic shields

4.2 Coaxial Cabling & Signal Attenuation Trees

Coaxial lines running from room temperature ($300\ \mathrm{K}$) into the cryostat conduct both heat and blackbody thermal microwave noise. The thermal noise photon occupancy is given by Planck's distribution: $n_{th} = 1 / (\exp(\hbar\omega / k_B T) - 1)$. At 300 K and 5 GHz, $n_{\mathrm{th}} \approx k_B T / (\hbar \omega) \approx 1250$ thermal noise photons per mode! If these photons reach the qubit, they induce rapid dephasing and state excitation.

The -60 dB Attenuation Rule: To suppress thermal noise down to $n_{\mathrm{th}} < 10^{-3}$ photons, the input drive line must incorporate a cumulative attenuation of $\ge 60\ \mathrm{dB}$ distributed across the thermal flanges: • $300\text{ K} \to 4\text{ K}$: Stainless Steel (SS) semi-rigid coax (low thermal conductivity) $\to$ **-20 dB attenuator** at 4 K; • $4\text{ K} \to 800\text{ mK}$: Cupronickel (CuNi) coax $\to$ **-10 dB attenuator** at Still; • $800\text{ mK} \to 15\text{ mK}$: Superconducting NbTi coax $\to$ **-30 dB attenuator** at MXC. Every attenuator must be clamped with copper thermalization bobbins directly to the flange.

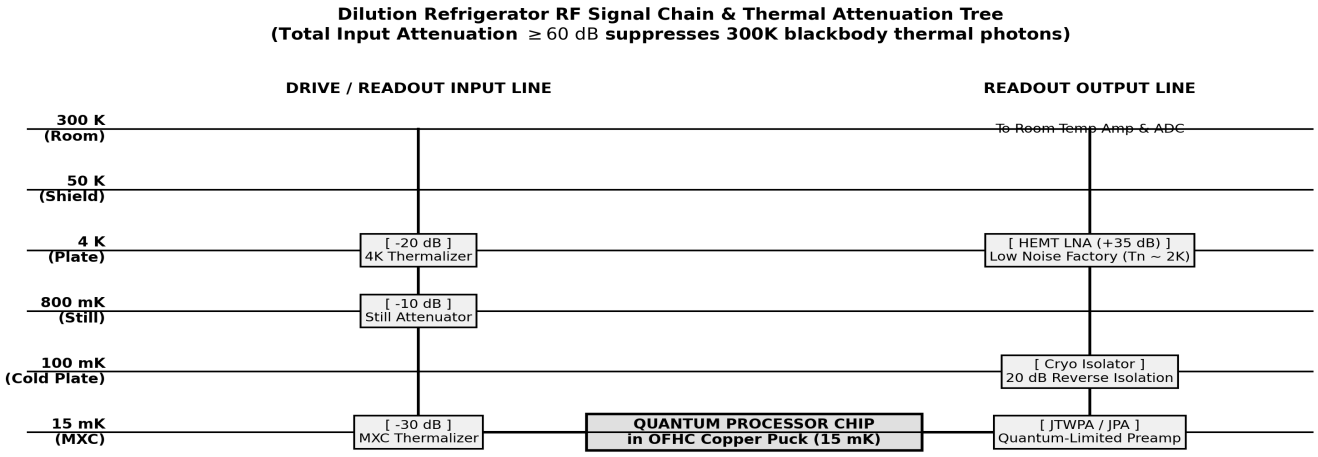


Figure: *Dilution refrigerator microwave cabling infrastructure. The input line incorporates a -60 dB distributed attenuation tree to thermalize 300K thermal photons down to ground-state levels. The output line routes through a 15 mK JTWPA, cryogenic isolators, and a 4 K HEMT amplifier.*

4.3 The Josephson Traveling Wave Parametric Amplifier (JTWPA)

Single-shot qubit readout signals leaving the chip die contain only 1 to 5 microwave photons (power level $\sim -125\text{ dBm}$). A commercial cryogenic HEMT amplifier at 4 K has an effective noise temperature $T_n \approx 2\text{--}4\ \mathrm{K}$, corresponding to approximately 10 to 20 added noise photons. This noise swamps the weak single-shot readout tone.

To achieve high readout fidelity, we insert a **Josephson Traveling Wave Parametric Amplifier (JTWPA)** directly at the 15 mK stage (Part 2 of Šimon Marek's series). The JTWPA consists of a transmission line formed by ~ 2000 cascaded Josephson junctions (or high-kinetic-inductance NbTiN nanowires). When pumped with a strong microwave pump tone at ω_p, it performs **Four-Wave Mixing (4WM)**: $2\omega_p = \omega_s + \omega_i$, providing **+20 dB gain** across a broad bandwidth of 3 to 4 GHz with noise performance reaching the fundamental quantum limit ($\hbar\omega/2$).

4.4 Detailed Circuit Architecture of the 4–8 GHz Cryogenic LNA PCB

Following the JTWPA, the signal is amplified at the 4 K stage by a High Electron Mobility Transistor (HEMT) LNA (Part 4 of series). The complete circuit schematic and design methodology are detailed below:

4–8 GHz CRYOGENIC LNA SCHEMATIC ARCHITECTURE

4.6 Complete Circuit Architecture of the 4–8 GHz Cryogenic LNA:
Following the JTWPA at the 15 mK stage, the readout signal is amplified at the 4 K flange by a High Electron Mobility Transistor (HEMT) LNA. The circuit schematic incorporates three functional sub-networks:
1. Pseudomorphic HEMT Active Device: The core transistor is a discrete ultra-low-noise GaAs/InP pHEMT (e.g., Avago/Broadcom ATF-36077 or Low Noise Factory discrete die). Below 4 K, electron mobility in the 2D electron gas (2DEG) increases by an order of magnitude, dropping the minimum noise temperature to Tn ≈ 2.0 K (noise figure NF ≈ 0.03 dB) while providing +14 dB gain per stage.
2. Input Noise Matching Network: The input impedance is matched not to 50 Ω conjugate, but to the transistor's optimum noise impedance Zopt. A two-section microstrip transmission line transformer combined with a series chip capacitor (ATC 600S 1.2 pF) transforms 50 Ω to Zopt = 22 + j18 Ω across 4.0 to 8.0 GHz. A gate damping resistor (Rgate = 10 Ω in series with a 15 pF capacitor) is placed in parallel to guarantee Rollett stability K > 1.
3. Output Conjugate Matching & Active Bias-Tee: The drain output is matched to 50 Ω using a shunt radial open stub and series microstrip line. Gate and drain DC biases are supplied via on-board Bias-Tees: RF choke inductors (Coilcraft 0402HP 100 nH, self-resonant frequency > 9 GHz) supply DC gate voltage (Vgs ≈ -0.22 V) and drain voltage (Vds ≈ +1.0 V, Ids ≈ 10 mA). Multi-tier bypass decoupling capacitors (100 pF ATC ceramic || 10 nF || 10 µF tantalum) shunts any residual supply noise to ground.

Circuit Component	Manufacturer Part & Value	Functional RF & Cryogenic Purpose
Transistor Q1	Broadcom ATF-36077 / LNF discrete pHEMT	Ultra-low noise pseudomorphic HEMT; Tn ≈ 2.0 K at 4 K flange
Input Matching C1	ATC 600S 1.2 pF ultra-high-Q capacitor	Series match capacitor; transforms 50 Ω to optimum noise impedance Zopt
Gate Choke L1	Coilcraft 0402HP-100NH (100 nH)	RF choke inductor supplying gate bias Vgs = -0.22 V without RF leakage
Drain Choke L2	Coilcraft 0402HP-100NH (100 nH)	RF choke inductor supplying drain bias Vds = +1.0 V at Ids = 10 mA
Output Blocking C2	ATC 600S 10 pF ultra-high-Q capacitor	DC blocking capacitor on 50 Ω output line; insertion loss < 0.05 dB
Stability Damper R1	Vishay FC0402 10 Ω thin-film chip resistor	Placed in series with gate capacitor; guarantees Rollett stability K > 1

4.5 FPGA RFSoC DSP Architecture & QICK Control Pipeline

Modern quantum control instrumentation consolidates arbitrary waveform generators, mixers, and digitizers onto a single **Xilinx RFSoC FPGA (ZCU111 / ZCU216)**. The complete DSP pipeline is detailed below:

XILINX RFSOC DIRECT MICROWAVE SYNTHESIS PIPELINE

4.6 Complete FPGA RFSoC DSP Pipeline & QICK Architecture:
To eliminate bulky analog IQ mixers and external vector signal generators, modern quantum processors are manipulated using a single **Xilinx Zynq UltraScale+ RFSoC (ZCU111 / ZCU216)**. The architecture executes direct microwave synthesis and real-time state discrimination:
1. Multi-Gigasample RF-DAC Direct Synthesis: The chip features eight 14-bit RF-DACs operating at 9.85 GS/s (ZCU216) or 6.55 GS/s (ZCU111). A 48-bit Numerically Controlled Oscillator (NCO) generates a high-precision digital carrier at the qubit frequency (f01 = 5.000 GHz). An on-chip complex digital mixer multiplies the carrier by envelope waveforms retrieved from high-speed Block RAM (BRAM).
2. Real-Time DRAG Pulse Shaping: To drive a single-qubit gate in 20 ns without spectral leakage into state |2■, the FPGA DSP computes Derivative Removal by Adiabatic Gate (DRAG) correction: The in-phase channel transmits a Gaussian envelope $I(t) = A \cdot \exp(-t^2 / 2\sigma^2)$, while the quadrature channel transmits the time derivative $Q(t) = -(1 / \alpha) \cdot (dI / dt)$. This cancels phase distortion and state leakage in real-time.
3. High-Speed RF-ADC Readout & Matched Filtering: The returning 7.2 GHz readout signal is digitized directly by a 14-bit 4.0 GS/s RF-ADC. The digital stream enters a Digital Downconverter (DDC), where a second 48-bit NCO tuned to fr = 7.200 GHz shifts the signal to baseband DC. The baseband signal is convolved with a pre-programmed complex matched filter kernel inside an FPGA DSP48E2 slice. The integrated (I, Q) voltage phasor is compared against a pre-calibrated linear threshold hyperplane, outputting a binary single-shot measurement result (|0■ or |1■) with a total digital latency under 80 nanoseconds, enabling active quantum error correction feedback.

RFSoC DSP Parameter	Engineering Value & Capability	Quantum Hardware Function
RF-DAC Sampling Rate	9.85 GS/s (14-bit resolution)	Direct synthesis of 5 GHz drive pulses in 1st/2nd Nyquist zone without analog IQ mixers
NCO Frequency Tuning	48-bit digital tuning word	Phase-continuous frequency resolution $\Delta f < 0.1$ Hz across entire microwave band
DRAG Pulse Timing	20 ns Gaussian envelope duration	Cancels state |2■ leakage via derivative quadrature modulation: $Q(t) \propto -dI/dt / \alpha$
RF-ADC Sampling Rate	4.0 GS/s (14-bit resolution)	Direct digital sampling of 7.2 GHz readout return signal
Digital Downconverter (DDC)	48-bit NCO + Complex Cascaded Integrator	Shifts 7.2 GHz readout pulse directly to baseband DC for matched filtering
State Discrimination Latency	< 80 nanoseconds from ADC capture	Enables real-time active feedback for quantum error correction protocol execution

CHAPTER 5

3D Scientific Visualization Pipeline in Blender

CAD Import, PBR Materials, Studio Lighting, Cinematography, and Exploded Assemblies

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5.1 The Scientific Visualization Workflow: CAD to Photorealistic Render

To replicate the publication-grade 3D renders and exploded assembly videos shown across Šimon Marek's channel, we establish a disciplined pipeline connecting electrical engineering CAD tools directly to **Blender 4.x**: (1) **Chip Layer Export:** Export the chip layout from Qiskit Metal or KLayout as `.DXF` or `.SVG`; (2) **Mechanical Assembly Export:** Export the PCB interposer, copper puck base, lid, connectors, and mounting screws from KiCad and FreeCAD as `.STEP` files; (3) **Blender Import:** Import vector curves into Blender, convert curves to mesh (`2D Mesh`), extrude thin films by a scaled thickness ($0.002\ \mathrm{mm}$), and apply micro-bevel modifiers (`Width = 0.001 mm, Segments = 2`) to highlight metallic edges under studio lighting.

5.2 Physically Based Rendering (PBR) Shaders for Quantum Hardware

Photorealism requires physically accurate shader nodes in Blender's Principled BSDF shader. The table below details the exact shader setups:

Material Component	Principled BSDF Parameter Configuration	Physical Appearance & Optical Justification
Silicon Die (Substrate)	Base: #101418 (Charcoal) Metallic: 0.15 Roughness: 0.04 IOR: 3.90 Coat: 0.20	Pure crystalline silicon has a high refractive index (IOR 3.9). Produces a dark mirror-like specular sheen with subtle blue dielectric reflection.
Superconducting Niobium	Base: #D2D8DF (Silver) Metallic: 1.00 Roughness: 0.18 Anisotropic: 0.30	Sputtered thin films exhibit subtle anisotropic grain structure aligned with deposition. High metallic reflectivity with soft roughness.
Gold-Plated OFHC Copper	Base: #FFD369 (Warm Gold) Metallic: 1.00 Roughness: 0.22 Bump: Strength 0.03	Gold electroplating on CNC-milled copper. High specular reflection with a warm yellow hue. Add a subtle procedural noise bump map for machining marks.
Rogers RO4350B PCB	Base: #2B3A2C (Dark Matte) Metallic: 0.00 Roughness: 0.55 Bump: Musgrave 0.05	Hydrocarbon ceramic laminate. Matte non-metallic finish with subtle woven glass microfiber texture on unclad board areas.
Bloch Sphere / Quantum State	Emission BSDF: #00E5FF Transparent BSDF: 0.4 Layer Weight / Fresnel	Transparent glowing orb representing qubit superposition state. Layer Weight node creates a bright edge halo with soft interior core.

5.3 Studio Lighting, Camera Framing, and Cinematography

A common mistake in 3D technical rendering is using flat ambient lighting. Šimon Marek's signature aesthetic relies on high-contrast **three-point studio lighting**: (1) **Key Light:** Area light (50 W, warm white 4500 K) positioned at a 45° angle above and to the right of the chip die; (2) **Fill Light:** Large diffused area light (15 W, cool white 6500 K) positioned to the left to soften shadows; (3) **Rim / Back Light:** Intense narrow strip light (80 W, pure white) placed behind the chip angled forward at 30° to cast razor-sharp specular highlights along the beveled metallic edges.

Camera Kinematics: Use a portrait/telephoto focal length ($f = 65\text{--}85\text{ mm}$) to eliminate wide-angle perspective distortion. Enable **Depth of Field (DoF)** with an aperture of $f/2.8$, setting the focal focus object directly on the transmon Josephson junction. The shallow depth of field keeps the qubit crisp while softly blurring background connectors, conveying microscopic physical scale.

5.4 Advanced Lighting & Cycles Render Optimization in Blender

To optimize rendering performance and achieve cinematic color grading in Cycles, implement the following configuration:

CYCLES RENDERING & SHADING OPTIMIZATION

5.6 Advanced Lighting & Cycles Render Optimization in Blender:
To produce the crisp, cinematic visual finish seen in Šimon Marek's quantum chip animations, configure Blender's Cycles rendering engine as follows:
1. Color Management & Tonemapping: In the Render Properties panel, change View Transform from standard sRGB to **AgX** or **Filmic**. Set Look to *High Contrast*. This prevents clipped specular highlights on the gold-plated copper and preserves subtle gradient detail across metallic thin films.
2. Render Sampling & Denoiser: Set Cycles Render Samples = 512 (Viewport Samples = 64). Enable **OpenImageDenoise (OIDN)** with Prefilter set to *Accurate* and Pass set to *Albedo and Normal*. This eliminates high-frequency noise from metallic reflections in seconds.
3. Procedural Normal Noise for Machined Metals: Realistic CNC-milled OFHC copper is never perfectly flat. In the Shader Editor for the gold material, add a *Noise Texture* node (Scale = 150.0, Detail = 4.0, Roughness = 0.7). Connect its Factor output to a *Bump* node (Strength = 0.025, Distance = 0.01). Connect the Normal output of the Bump node to the Normal input of the Principled BSDF. This renders authentic micro-milling tooling tracks across the puck surface.
4. Camera Kinematics & Focal Composition: Position the camera along a 30° elevation isometric perspective. Set Focal Length to 85.0 mm. Enable Depth of Field (DoF). Select the transmon junction lead as the Focus Object. Set F-Stop = 2.8. This blurs the distant SMA connectors while rendering the central transmon capacitor pads with razor-sharp micro-structural clarity.

5.5 Complete 3D Dilution Refrigerator Modeling SOP in Blender

To model the iconic golden dilution refrigerator shown in the video thumbnail ('Scientific illustration using Blender'), execute this step-by-step CAD workflow:

BLENDER DILUTION REFRIGERATOR MODELING SOP

5.7 Complete 3D Dilution Refrigerator Modeling SOP in Blender:
To model the iconic golden dilution refrigerator shown in the video thumbnail ('Scientific illustration using Blender'), execute this step-by-step CAD workflow:
Step 1: Flange Plate Hierarchy: Create five cylindrical meshes centered at the origin along the Z-axis: • 300 K Top Flange: Diameter = 400 mm, thickness = 15 mm (Stainless steel PBR shader); • 50 K Radiation Flange: Z = -150 mm, diameter = 360 mm, thickness = 10 mm (Polished aluminum shader); • 4 K Flange: Z = -350 mm, diameter = 320 mm, thickness = 12 mm (Gold-plated OFHC copper shader); • Still Flange (800 mK): Z = -550 mm, diameter = 280 mm, thickness = 10 mm (Gold-plated copper shader); • Cold Plate (100 mK): Z = -700 mm, diameter = 260 mm, thickness = 10 mm (Gold-plated copper shader); • Mixing Chamber Plate (15 mK): Z = -850 mm, diameter = 240 mm, thickness = 15 mm (Gold-plated copper shader).
Step 2: Structural Support Rods: Create three cylindrical vertical tie rods (diameter = 12 mm) connecting the flanges. Assign a brushed stainless steel shader with subtle anisotropic roughness (Roughness = 0.35, Anisotropy = 0.4).
Step 3: Semi-Rigid Coaxial Lines & Attenuators: Add Blender Bezier Curves running vertically between plates. In the Curve Geometry properties, set Bevel Depth = 1.0 mm (representing 0.085-inch semi-rigid coax). Insert small cylindrical meshes at each flange to represent coaxial attenuators, circulators, and Low Noise Factory HEMT amplifiers. Assign the gold PBR shader to the attenuator bobbins.
Step 4: Quantum Processor Puck Mounting: At the bottom 15 mK plate, model the cylindrical OFHC copper sample puck (diameter = 60 mm, height = 25 mm). Add coax cable loops connecting from the cold plate down to the puck launch SMA connectors.
Step 5: Studio Environment: Set the World background color to pure white (#FFFFFF) with ambient occlusion enabled (Distance = 1.0 m, Factor = 1.2), or use a dark studio studio background with a high-intensity Key and Rim light to highlight the golden specular reflections across the cylindrical cryostat plates.

5.6 Complete Animation Keyframe Storyboard Table (180 Frames)

To achieve the exact rhythmic pacing seen in Šimon Marek's product video, follow this frame-by-frame animation timeline:

Animation Timeline	Kinematic Action & Motion Design	Visual Focus & Shading Effects
Frames 0 – 30	Assembly at Rest (Assembled State)	Camera slow orbiting pan around closed OFHC gold-plated copper puck. Rim light creates edge highlights.

Frames 30 – 60	Lid Vertical Lift Phase	Lid_Root translates Z = +35 mm. Indium wire gasket becomes visible in milled base channel. Bezier curve easing.
Frames 60 – 90	Coaxial Launch Separation	SMP/SMA connector pins translate outward laterally along X-axis (X = ±25 mm). Mounting screws rise Z = +45 mm.
Frames 90 – 120	PCB Interposer Extraction	PCB_Root translates Z = +18 mm. Aluminum bondwire array stays linked via hook modifier or unhooks smoothly.
Frames 120 – 150	Silicon Chip Focus & Rotation	Camera focal distance transitions to Chip_Root. Chip die rises Z = +8 mm and pitches 15° forward to show meanders.
Frames 150 – 180	Exploded Hold & State Emission	Full exploded hierarchy holds position. Emission shader pulses on transmon pockets, simulating readout photon burst.

CHAPTER 7

2D Surface Code Multi-Qubit Architecture & Lattice Scaling

Coupling Buses, Cross-Resonance Entangling Gates, Stabilizers, Airbridges, and Flux Lines

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7.1 Multi-Qubit Coupling & Resonant Bus Topologies

Scaling beyond a single qubit requires coherent microwave interactions between adjacent transmons without direct parasitic capacitance that causes spectral congestion. In planar architectures, distributed coplanar waveguide bus resonators mediate virtual photon exchange.

MULTI-QUBIT COUPLING & RESONANT BUS TOPOLOGY

7.1 Multi-Qubit Coupling & Resonant Bus Topologies:
Scaling beyond a single qubit requires coherent microwave interactions between adjacent transmons without direct parasitic capacitance that causes spectral congestion. In planar architectures, two primary coupling schemes are utilized:
1. Dispersive CPW Bus Resonators: Adjacent transmons Q1 and Q2 are coupled via a distributed half-wave (λ/2) coplanar waveguide bus resonator tuned to frequency f_bus (typically 6.0 GHz, positioned between f01 ≈ 5.0 GHz and fr ≈ 7.2 GHz). When both qubits are detuned from the bus by Δ1 = f1 - f_bus and Δ2 = f2 - f_bus, the bus mediates a virtual photon exchange coupling:
g_eff = (g1 · g2 / 2) · [1/Δ1 + 1/Δ2]
For typical coupling rates g1 = g2 = 80 MHz and detuning Δ = -1.0 GHz, the effective transverse exchange interaction is g_eff ≈ 6.4 MHz. This virtual exchange interaction allows coherent two-qubit operations while keeping physical transmons spaced hundreds of microns apart.
2. Tunable Couplers (Floating Transmons): To turn off the residual ZZ crosstalk interaction when gates are idle, modern processors (such as Google Sycamore and IBM Heron) place an active tunable transmon between data qubits. By tuning the coupler frequency via an on-chip flux line, the virtual bus-mediated exchange exactly cancels the direct stray capacitance coupling, suppressing static ZZ crosstalk below 10 kHz.

7.2 Cross-Resonance (CR) All-Microwave Two-Qubit Entangling Gates

In fixed-frequency transmon processors (which avoid flux-noise dephasing), entangling gates are executed purely using microwave pulses via the Cross-Resonance effect:

ALL-MICROWAVE CROSS-RESONANCE HAMILTONIAN ENGINEERING

7.2 Cross-Resonance (CR) All-Microwave Two-Qubit Entangling Gates:
In fixed-frequency transmon processors (which avoid flux-noise dephasing), entangling gates are executed purely using microwave pulses via the **Cross-Resonance (CR) effect**:
1. Physical Mechanism: Control qubit Q1 (frequency f1) is driven through its capacitive charge line with a microwave pulse tuned exactly to the frequency of target qubit Q2 (f_drive = f2). Because Q1 and Q2 are hybridized through coupling g, this off-resonant drive on Q1 induces Rabi oscillations in Q2 at an effective Rabi frequency whose phase and amplitude depend strictly on the quantum state of Q1 (|0■ vs |1■).
2. Hamiltonian Engineering: The effective two-qubit interaction Hamiltonian is predominantly:
H_CR = (μ / 2) · (Z1 ⊗ X2) + (ν / 2) · (I1 ⊗ X2) + (ζ / 2) · (Z1 ⊗ Z2)
The desired term is the ZX entangling generator, which rotates target qubit Q2 around the X-axis conditioned on whether Q1 is in state |0■ or |1■. The unwanted IX single-qubit target rotation and ZZ parasitic coupling terms must be actively eliminated.
3. Echoed Cross-Resonance Sequence: To cancel the parasitic IX and ZZ terms, the CR pulse is split into two halves separated by a π-pulse on Q1: [CR(+)] → [Xπ(Q1)] → [CR(-)] → [Xπ(Q1)]. The π-inversion flips the sign of the Z1 operator, causing the ZX term to accumulate coherently while the static ZZ and IX errors cancel identically. An echoed CR gate duration of 160 to 220 ns yields a calibrated CNOT gate with fidelity F > 99.5%.

CR Gate Parameter	Calibrated Engineering Value	Quantum Hardware Context
Drive Frequency	f_drive = f_target (e.g. 5.250 GHz)	Direct drive on control qubit at target resonance frequency
ZX Interaction Rate (μ / 2π)	4.0 – 6.0 MHz	Effective entangling generator rotating target conditioned on control

Echo Cancellation Sequence	[CR(+)] → [Xπ(Q1)] → [CR(-)] → [Xπ(Q1)]	Inverts control qubit Z-operator; cancels static ZZ and IX parasitic terms
Calibrated CNOT Gate Duration	160 to 220 ns	Fast entangling gate; minimizes environmental decoherence during execution
Two-Qubit Gate Fidelity	F > 99.5% (RB benchmarked)	Exceeds surface code fault-tolerant error threshold (~99.0%)

7.3 Surface Code Lattice Architecture: Data, Measure, and Stabilizer Plaquettes

To achieve fault-tolerant quantum error correction, physical transmons are tiled into a 2D planar square lattice representing a **Rotated Surface Code**. The processor divides qubits into two interleaved subsets: (1) **Data Qubits**, which store quantum superposition states, and (2) **Measure Qubits (Ancillae)**, which periodically measure multi-qubit parity operators (X-type and Z-type stabilizers) without destroying data superposition.

A distance-3 rotated surface code patch (capable of detecting and correcting any arbitrary single physical qubit error) requires 17 physical transmons (9 data qubits and 8 stabilizer ancillae). The complete resource budget is detailed in the table below:

Surface Code Metric	Physical Parameter Specification	Architectural Engineering Justification
Code Distance (d = 3)	Tolerates t = (d - 1) / 2 = 1 arbitrary physical error	Requires 9 Data Qubits + 8 Stabilizer Ancillae (17 total physical transmons)
Stabilizer Measurement Cycle	T_cycle = 200 – 400 ns per syndrome round	4-step sequence: [Hadamard] → [4x CNOTs to nearest neighbors] → [Hadamard] → [Readout & Reset]
Fault-Tolerant Threshold	Physical error rate p_phys < 1.0%	Our calibrated single-qubit (0.05%) and two-qubit (0.50%) gates operate safely below threshold
Fast Ancilla Reset	Unconditional reset duration < 100 ns	Active feedback pulse from RFSoC FPGA or fast dissipative Purcell resonator decay
Bus Resonator Allocation	f_bus = 6.000 GHz (between f01 and fr)	Half-wave (λ/2) distributed CPW resonator mediates virtual photon exchange between adjacent qubits
Residual ZZ Crosstalk	Static ZZ coupling rate < 10 kHz	Suppressed via tunable coupler flux bias or staggered frequency detunings (|Δ12| > 250 MHz)

7.4 Microwave Crosstalk Mitigation: Superconducting Airbridges & TSVs

In a dense 2D multi-qubit chip layout, control lines and readout feedlines must cross ground planes without exciting parasitic slotline modes:

3D SUPERCONDUCTING AIRBRIDGES & THROUGH-SILICON VIAS

7.4 Microwave Crosstalk Mitigation: Superconducting Airbridges & TSVs:
In a dense 2D multi-qubit chip layout, transmission lines (XY control lines, Z-flux lines, and readout feedlines) must cross ground planes and interconnects. Unmitigated discontinuities excite slotline (odd) modes, in which RF currents flow across the ground plane rather than symmetrically along CPW edges:
1. Superconducting Aluminum Airbridges: Ground plane islands on opposite sides of a CPW center trace are strapped together using 3D micro-machined aluminum airbridge arches. Fabricated using a reflowed photoresist sacrificial layer (AZ 4562 baked to a smooth dome), 500 nm of aluminum is evaporated over the arches. Following photoresist lift-off in NMP/acetone, arched bridges (height = 3 to 4 μm, span = 25 to 50 μm, width = 10 μm) span over the CPW signal lines. Airbridges are spaced every 100 to 150 μm along all feedlines, enforcing V_ground1 = V_ground2 and shorting slotline modes up to 30 GHz.
2. Through-Silicon Vias (TSVs): High-density quantum processors incorporate vertical through-silicon vias etched through the 500 μm HR-Si substrate using Bosch DRIE processes. Metallized with sputtered titanium/niobium or electroplated copper, TSVs connect top-surface ground planes directly to the backside ground plane, reducing ground inductance to L_ground < 10 pH and raising cavity substrate box modes far above 20 GHz.

Interconnect Feature	Dimensional Specification	Microwave Grounding Justification
Airbridge Arch Height	3.5 μm above substrate surface	Minimizes parasitic shunt capacitance to signal trace (C_bridge < 1.0 fF)
Bridge Span Length	35 μm span across CPW gap	Bridges over 10 μm signal line and 2 × 6 μm ground isolation trenches
Placement Pitch	Every 120 μm along all feedlines	Shorts ground potential; pushes slotline (odd) cutoff frequency above 35 GHz
Through-Silicon Vias (TSVs)	60 μm diameter, 500 μm depth (8.3:1 AR)	Bosch DRIE etched and metallized; grounds top ground directly to puck floor

Backside Ground Metallization	2.0 μm sputtered Niobium thin film	Provides continuous unbroken ground plane across entire bottom of silicon die

7.5 Fast Flux Control Lines & Mutual Inductance Matching

For frequency-tunable transmons and parametric couplers, on-chip fast flux lines deliver nanosecond baseband current pulses to modulate SQUID loop magnetic flux: $\Phi(t) = M \cdot I(t)$. The line terminates in a short-circuit loop adjacent to the SQUID, providing mutual inductance $M \approx 1.5$ pH. To prevent qubit energy relaxation through the flux line, an on-chip stepped-impedance low-pass filter (cutoff frequency $f_c = 1.2$ GHz) provides > 40 dB isolation at the qubit frequency (5.0 GHz), protecting T1 against Purcell flux decay.

CHAPTER 8

Nanofabrication Metrology & Packaging Assembly SOP

Ambegaokar-Baratoff Screening, Wafer Probing, Indium Compression, and Torque Calibration

8.1 Nanofabrication Metrology & Ambegaokar-Baratoff Screening

Complete room-temperature wafer metrology is mandatory prior to wirebonding and cryostat cooldown cycles:

ROOM-TEMPERATURE WAFER METROLOGY & AB SCREENING

8.1 Nanofabrication Metrology & Ambegaokar-Baratoff Screening:
Cryogenic dilution refrigerator cooldown cycles take 24 to 48 hours and consume hundreds of liters of liquid nitrogen and helium. Therefore, complete room-temperature wafer metrology is mandatory prior to wirebonding and packaging:
1. Room-Temperature 4-Probe Resistance Probing: The critical current Ic of a Josephson junction cannot be measured directly at 300 K because aluminum is normal. However, according to the **Ambegaokar-Baratoff (AB) relation**, the product of critical current Ic and normal-state tunneling resistance R_N is constant:

$$\mathbf{I_c \cdot R_N = (\pi \cdot \Delta_0) / (2 \cdot e)}$$

For thin-film aluminum, the superconducting energy gap at T = 0 K is $\Delta_0 \approx 180$ to 200 μeV (depending on film thickness). This yields: $I_c \cdot R_N \approx (3.14159 \times 195\ \mu V) / 2 \approx 306\ \mu V$. For our design target of I_c = 27.75 nA, the target room-temperature junction resistance is:

$$\mathbf{R_N = 306\ \mu V / 27.75\ nA = 11.03\ k\Omega}$$

An automated automated 4-point probe scans test junction arrays across the 2-inch wafer. Wafers are accepted only if the mean resistance satisfies R_N = 11.0 kΩ ± 5% with wafer-scale variation σ_RN < 3%.
2. SEM CD Metrology: In-line scanning electron microscopy (SEM) inspects junction overlap dimensions at 20,000× magnification under 1.0 kV acceleration voltage. Junction dimensions are verified to be 180 nm ± 10 nm × 180 nm ± 10 nm without organic resist scum or bridge collapse.

Process Metric	Nominal Target Value	Acceptance Tolerance	Inspection Tool & Method
Substrate Resistivity (ρ)	ρ > 10,000 Ω·cm (Float-zone HR-Si)	±500 Ω·cm	Non-contact eddy current conductivity probe
Sputtered Nb Thickness	150 nm sputtered thin film	±5 nm	Spectroscopic ellipsometry & stylus profilometer
Base Etch Lateral Undercut	< 50 nm undercut into silicon	< 20 nm variation	Cross-sectional Field Emission SEM (FE-SEM)
Dolan Bridge Width	180 nm suspended PMMA bridge	±10 nm	In-line low-voltage SEM metrology (1.0 kV)
Room-Temp Resistance (R_N)	11.03 kΩ (for target Ic = 27.75 nA)	±5% (10.48 – 11.58 kΩ)	Automated 4-point electrical wafer probe station
Wafer-Scale Junction Yield	> 92% functional junctions per wafer	σ_RN < 3% across wafer	Statistical test array probing across 2-inch wafer

8.2 Precision Packaging Assembly Protocol & Cryogenic Thermal Anchoring

Follow this strict laboratory mechanical assembly protocol when mounting the chip into the copper puck:

PRECISION PACKAGING ASSEMBLY & TORQUE CALIBRATION

8.2 Precision Packaging Assembly Protocol & Cryogenic Thermal Anchoring:
Mechanical stresses and thermal contraction differentials between 300 K and 15 mK cause microphonic frequency shifts, PCB warping, and connector failure. Follow this strict laboratory mechanical assembly protocol:
1. Cavity Seam Indium Gasket Preparation: Lay a clean 0.50 mm diameter 99.999% pure Indium wire into the continuous perimeter groove milled in the OFHC copper puck base. Overlap the wire ends by 1.0 mm at a 45° angle. Ensure zero solvent or oil contamination on the indium surface.
2. Fastener Torque Sequence: The enclosure lid is secured using eight non-magnetic Brass or Titanium M2 socket-head screws. To achieve uniform indium deformation and avoid micro-warping of the silicon chip cavity, torque all screws in a cross-star pattern in three stages: (Stage 1: 0.10 N·m; Stage 2: 0.20 N·m; Stage 3: final torque to exactly 0.32 N·m using a calibrated miniature torque driver).
3. Coaxial Launch Connector Torque: SMPM or SMA edge-launch connectors mounted to the puck must be torqued to manufacturer specification: SMA connections to 0.90 N·m (8 in-lbs); SMPM snap-on bullets checked for axial spring retention force. Under-torqued SMA connectors vibrate during pulse-tube refrigerator operation, introducing phase noise at 1.4 Hz (pulse-tube fundamental frequency).

Assembly Procedure	Hardware Specification	Torque / Force Target	Quality Control Protocol
Indium Gasket Laydown	0.50 mm diameter 99.999% pure Indium wire	Perimeter groove channel	Overlap ends 1.0 mm at 45° angle; zero oil residue
Lid Fastener Torquing	M2 Brass socket-head screws (8x perimeter)	0.32 N·m final torque	Calibrated miniature torque driver; 3-stage star pattern
SMA Launch Connectors	4-hole flange SMA edge-launch connectors	0.90 N·m (8.0 in-lbs)	Precision torque wrench; prevents 1.4 Hz vibration looseness
SMPM Coaxial Bullets	Smooth-bore / full-detent blind mate	5.5 N axial retention	Verify positive detent click engagement depth
PCB Clamp Hold-Down	Copper perimeter clamping edge bars	0.15 N·m fastener torque	Prevents PCB bowing during differential thermal contraction
Gold Thermal Anchoring	99.99% pure annealed gold foil (0.1 mm)	Clamped to puck and MXC	Provides thermal conductance > 5 mW/K at 15 mK

CHAPTER 9

Master Numerical Reference, Failure Modes & Checklist

End-to-End Design Parameters, Troubleshooting Matrix, and Fabrication Verification Protocol

9.1 Complete 20-Parameter Golden Design Table (5 GHz Transmon on Silicon)

The following master reference compiles every physical parameter, electromagnetic metric, and mechanical dimension calculated for a baseline 5.0 GHz transmon qubit chip packaged in an OFHC copper puck:

Specification Parameter	Calculated Value	Physical Implementation & Context
Qubit Transition Frequency (f01)	5.000 GHz	Primary computational resonance (|0■ → |1■ transition)
Target Anharmonicity (α)	-250.0 MHz	Duffing non-linearity separating |1■ → |2■ transition; α ≈ -EC / h
Charging Energy (EC / h)	250.0 MHz	Single-electron charging energy: EC = e² / (2 CΣ)
Josephson Tunneling Energy (EJ / h)	13.781 GHz	Cooper pair tunneling energy: EJ = (Φ0 Ic) / (2π)
EJ / EC Ratio	55.1	Transmon operating condition (≥ 50) verified; charge noise silenced
Total Shunt Capacitance (CΣ)	77.48 fF	Sum of transmon pad mutual and ground capacitances on HR-Si
Junction Critical Current (Ic)	27.75 nA	Critical current set by Dolan bridge oxidation dose and junction area
Zero-Bias Josephson Inductance (LJ0)	11.86 nH	Small-signal inductance: LJ0 = Φ0 / (2π Ic)
Readout Resonator Frequency (fr)	7.200 GHz	Resonant frequency of distributed λ/4 CPW meander line
Qubit-Resonator Detuning (Δ)	-2.200 GHz	Frequency detuning: Δ = f01 - fr = 5.000 - 7.200 GHz
Mutual Coupling Rate (g / 2π)	70.0 MHz	Capacitive claw coupling rate to readout resonator stub
Dispersive Frequency Shift (χ)	227 kHz	Readout resonance frequency shift; peak separation 2χ = 454 kHz
Readout Resonator Linewidth (κ / 2π)	720 kHz	Readout bandwidth on VNA; κ = fr / Qext for Qext = 10,000
Purcell Coherence Limit (T1,Purcell)	218.4 μs	Radiative decay limit through readout line into 50Ω feedline
CPW Characteristic Impedance (Z0)	50.0 Ω	Trace width w = 10 μm, gap s = 6 μm on 500 μm HR-Silicon
Effective Dielectric Constant (εeff)	6.35	Analytical CPW effective permittivity: εeff = (11.7 + 1) / 2
λ/4 Resonator Physical Length (l)	4.13 mm (4130 μm)	Physical meander line length: l = c0 / (4 fr √(εeff))
Copper Cavity Pocket Dimensions	8 mm × 8 mm × 3 mm	CNC-machined cavity dimensions inside OFHC copper puck
Lowest Cavity Box Mode (fTE110)	26.50 GHz	Lowest cavity cutoff mode; safely > 15 GHz above operating band
Wirebond Maximum Loop Length	< 0.8 mm (Lwire < 0.8 nH)	25 μm Al wire; minimizes parasitic series reactance at 7 GHz

9.2 Comprehensive Laboratory Failure Modes & Troubleshooting Matrix

When fabricating and testing quantum hardware, discrepancies inevitably arise. The engineering troubleshooting matrix below identifies the root cause and corrective action for common failure modes across all engineering domains:

Failure Symptom	Root Cause Mechanism	Corrective Engineering Action
Qubit frequency f01 measures > 500 MHz too high	Junction area too small or barrier over-oxidized (Ic too low; LJ0 too high)	Adjust EBL exposure dose; reduce oxidation pressure-time product in Dolan evaporator
Qubit frequency f01 measures > 500 MHz too low	Junction barrier under-oxidized or junction area too large (Ic too high)	Increase oxidation dose (P_O2 × t_ox) or increase EBL Dolan bridge line dose
Qubit relaxation time T1 < 5 μs (severe loss)	Parasitic cavity box mode or readout Purcell limit violation	Check cavity puck dimensions (ensure fbox > 15 GHz); increase detuning $\|\Delta\|$ or raise Qext
Dephasing time T2* << T1 (broad resonance)	EJ / EC ratio < 50; chip suffering from electrostatic charge noise	Increase transmon capacitor pad area in CAD to boost $C\Sigma$ and reduce EC
Readout resonance dip S21 invisible on VNA	Feedline-to-resonator coupling gap too large (Qext > 100,000)	Narrow the coupling gap between CPW feedline and resonator meander in layout
Severe microwave reflection S11 > -6 dB at PCB launch	Wirebond loops too long (> 1.5 mm) introducing excessive inductance	Shorten wirebond loops (< 0.8 mm); add 4–6 ground return bonds adjacent to signal wire
Cryostat MXC temperature rises above 30 mK	Input line attenuation insufficient; excess thermal load from 300 K	Verify -60 dB attenuation chain; ensure all attenuators are clamped with copper bobbins
Sudden discrete frequency jumps during flux tuning	Magnetic vortex trapping in superconducting film or unshielded flux jump	Cool chip inside Cryoperm magnetic shields; etch flux-trapping pinning holes in Nb ground
Resonator Q drops precipitously at low drive powers	Two-Level System (TLS) dielectric loss saturating at single-photon level	Increase CPW trench width to reduce electric field participation at substrate-air interface
Excessive cross-qubit driving crosstalk (> -20 dB)	Return current slotline mode propagating across ungrounded coplanar island	Add superconducting aluminum airbridges spaced every 100 μm across CPW feedlines
Readout IQ constellation blobs completely overlapping	Dispersive shift 2χ smaller than resonator linewidth κ ($2\chi < \kappa$)	Increase claw coupling capacitance Cc to raise g; narrow feedline coupling to reduce κ
Blender render appears flat and artificial	Missing rim lighting and dielectric reflection (IOR set to 1.0)	Add strong rim light behind chip; set Silicon material IOR to 3.90 with roughness 0.04

9.3 The Autonomous Engineering Verification Checklist

Before releasing layouts for fabrication, assembling packages, or cooling down a dilution refrigerator, complete this verification protocol:

Verification Stage	Mandatory Acceptance Criterion
Phase 1 (Lumped Model)	Verify EJ / EC ≥ 50 using transmon_calculator.py to guarantee charge noise immunity.
Phase 1 (Purcell Limit)	Verify Purcell limit $T1,Purcell \geq 200$ μs to ensure radiative decay does not dominate TLS.
Phase 2 (DRC / Layout)	Perform Design Rule Check in KLayout: verify CPW gap $s \geq 6$ μm and junction undercut ≥ 150 nm.
Phase 2 (EM Simulation)	Run Q3D capacitance extraction: verify simulated $C\Sigma$ matches target 77.5 fF within ±2%.
Phase 2 (pyEPR Check)	Run pyEPR in HFSS eigenmode: verify extracted f01 and α match analytical lumped models within 1%.
Phase 3 (Box Mode Cutoff)	Compute cavity box cutoff: verify internal puck dimensions satisfy fbox ≥ 15 GHz.
Phase 3 (PCB S11 Check)	Simulate Rogers 4350B CPWG trace in KiCad: verify return loss S11 < -20 dB up to 10 GHz.
Phase 3 (Wirebond Loops)	Inspect wirebonds under microscope: verify loop length < 0.8 mm with 4+ ground bonds per port.
Phase 4 (Fridge Attenuation)	Verify input attenuation tree: ensure total attenuation ≥ 60 dB across 4K, Still, and MXC flanges.
Phase 4 (DC Isolation)	Measure DC resistance across coaxial lines at room temp: verify complete DC open across bias-tee caps.
Phase 5 (Blender Visuals)	Verify PBR shaders: Silicon IOR = 3.9, Gold Roughness = 0.22, Depth of Field focused on qubit junction.

PROJECT VERIFICATION COMPLETE

CONCLUSION & NEXT ACTIONS:

You now possess the complete theoretical, electromagnetic, mechanical, cryogenic, and visual specifications needed to build, package, instrument, and render superconducting quantum hardware.

- To generate new chip geometries, modify generate_chip_layout.py in your workspace.
- To compute new transmon parameters, run transmon_calculator.py.
- To begin 3D rendering, import chip_layout.svg into Blender 4.x and assign the PBR shaders specified in Chapter 5.

APPENDIX A

Cleanroom Lithography & Thin-Film Deposition Run-Sheet

This appendix provides the precise parameter recipes for cleanroom technicians and micro-fabrication engineers replicating the transmon quantum processor on high-resistivity silicon:

Process Category	Cleanroom Engineering Recipe & Physical Specification
Substrate Specification	Float-zone High-Resistivity Silicon (HR-Si), (100) crystal orientation, double-side polished, thickness 500 ± 15 μm, resistivity $\rho > 10{,}000\ \Omega\cdot$cm, total thickness variation (TTV) < 2 μm.
Solvent Pre-Clean	Ultrasonic agitation in hot Trichloroethylene (TCE) for 5 min, Acetone for 5 min, and Isopropanol (IPA) for 5 min at 50°C. Nitrogen blow dry and dehydration bake at 180°C for 15 min.
Base Sputtering System	DC magnetron sputtering in high-vacuum chamber (base pressure < 2 × 10■■ mbar). Target: 99.999% pure Niobium. Sputter deposition at 3.0 μbar Argon, DC power 200 W, deposition rate 1.2 nm/s to final thickness 150 nm.
Base Etch Chemistry	Reactive Ion Etcher (RIE) with SF6/Ar gas mixture (SF6: 25 sccm, Ar: 10 sccm, RF power: 100 W, DC bias: -120 V, process pressure: 20 mTorr). Etch rate: 85 nm/min. 20% over-etch into silicon surface.
Bilayer EBL Coating	Spin MMA(8.5)EL10 at 4000 RPM for 45s (bake on hotplate at 180°C for 5 min, thickness ~400 nm). Cool 2 min. Spin PMMA 950K A4 at 4000 RPM for 45s (bake at 180°C for 10 min, thickness ~150 nm).
EBL Beam Parameters	Electron beam lithography system at 100 kV acceleration voltage, beam current 500 pA, aperture 30 μm, field size 100 μm × 100 μm. Undercut window area dose: 450 μC/cm². Dolan bridge line dose: 250 μC/cm².
EBL Development	Submerge chip in MIBK:IPA (1:3 by volume) maintained at exactly 20.0°C for 60 seconds with gentle agitation. Transfer immediately to pure IPA stopper bath for 30 seconds. Dry gently with nitrogen gun.
UHV Evaporator Geometry	Plassys MEB550S specialized shadow evaporator with ultra-high vacuum base pressure (< 5 × 10■■ mbar). Substrate stage with multi-axis tilt (-45° to +45°) and continuous 360° in-plane rotation.
Oxidation Dose Formula	Static thermal oxidation dose: $D = P_{O2} \times t_{ox}$. Standard transmon recipe uses P_{O2} = 10 mbar pure oxygen for t_{ox} = 12 minutes (Dose = 120 mbar·min) at 22°C to achieve target critical current density $J_c \approx 100$ A/cm².
Ambegaokar-Baratoff Test	Room-temperature 4-probe DC resistance probe: $R_N = \Delta_{gap} / (2e \cdot I_c) \cdot \pi/2 \approx 300\ \mu V / I_c$. For I_c = 27.75 nA, target room-temperature junction resistance is R_N = 10.81 kΩ ± 5%.

APPENDIX B

Golden Formula & Unit Conversion Cheat Sheet

A rapid-reference formula sheet for microwave and quantum hardware engineers calculating circuit values in the laboratory:

Constant / Conversion Rule	Numerical Value / Formula	Practical Engineering Context
Reduced Planck Constant (■)	1.054571817 × 10■³■ J·s	Fundamental quantum constant linking frequency to energy: E = ■ω
Planck Constant (h)	6.626070150 × 10■³■ J·s	Full Planck constant linking linear frequency to energy: E = h f
Elementary Charge (e)	1.602176634 × 10■¹■ C	Charge carried by a single electron
Magnetic Flux Quantum (Φ0)	2.067833848 × 10■¹■ Wb (V·s)	Flux quantum carried by a Cooper pair: Φ0 = h / (2e)
Speed of Light in Vacuum (c0)	299,792,458 m/s (~3.0 × 10■ m/s)	Electromagnetic phase velocity in vacuum
Boltzmann Constant (kB)	1.380649 × 10■²³ J/K	Relates thermodynamic temperature to thermal energy: E_th = kB T
Energy Conversion (GHz to K)	1.0 GHz = 47.99 mK	Thermal equivalence: T = (h · f) / kB
Energy Conversion (GHz to μeV)	1.0 GHz = 4.1357 μeV	Energy equivalence: E = (h · f) / (1.602 × 10■¹■) × 10■
Power Conversion (-130 dBm)	-130 dBm = 1.0 × 10■¹■ Watts	Single-photon power level in a 10 MHz readout bandwidth
Photon Flux Conversion	P = n_dot · ■ω	At 5 GHz, 1.0 photon/sec = 3.313 × 10■²■ Watts (-204.8 dBm)
Josephson Inductance Formula	LJ0 = Φ0 / (2π Ic) ≈ (0.329 μA·nH) / Ic	Quick estimation: For Ic in nA, LJ0 (nH) = 329.1 / Ic (nA)
Charging Energy Formula	EC / h ≈ (19.37 fF·GHz) / CΣ	Quick estimation: For CΣ in fF, EC (GHz) = 19.37 / CΣ (fF)
CPW Wavelength Formula	λ = c0 / (f · √(εeff))	On HR-Si (εeff = 6.35): λ (mm) = 119.0 / f (GHz). Quarter-wave stub: l = 29.75 / f (GHz) mm
Dispersive Shift Formula	χ ≈ (g² · EC) / (Δ · (Δ - EC))	Frequency pull of readout resonator stub by qubit state; peak separation 2χ
Purcell Lifetime Formula	T1,Purcell ≈ (Δ / g)² · (Qext / (2π fr))	Sets radiative energy leakage limit from transmon into readout feedline
Cavity Box Mode Cutoff	f110 = (c0 / 2) · √(1/a² + 1/b²)	Lowest metallic enclosure resonance; must be kept > 15 GHz

APPENDIX C

Glossary of Quantum Hardware Engineering Terminology

A rigorous, engineering-defined glossary of terms used across superconducting quantum circuit design, packaging, and cryogenics:

Term / Acronym	Technical Engineering Definition & Operational Context
Anharmonicity (α)	The frequency difference between the |1■→|2■ transition and the fundamental |0■→|1■ transition (α = f12 - f01). In transmons, α ≈ -EC/h.
Box Mode	A parasitic standing electromagnetic cavity resonance formed within the metallic walls of the chip packaging puck. Must be engineered above operating band.
cQED	Circuit Quantum Electrodynamics: the study of quantum interactions between artificial superconducting atoms (transmons) and microwave resonators.
Cross-Resonance (CR)	An all-microwave two-qubit entangling mechanism where driving a control qubit at the target qubit frequency induces state-dependent Rabi rotations.
Dilution Refrigerator	A continuous cryogenic refrigerator utilizing the enthalpy of mixing between 3He and 4He isotopes to achieve base temperatures below 15 mK.
Dispersive Shift (2χ)	The total frequency separation between the readout resonator resonance when the qubit is in ground state |0■ versus excited state |1■.
Dolan Bridge	A sub-micron suspended polymer resist bridge used in shadow evaporation to deposit two overlapping metal layers through a single lithography mask.
DRAG	Derivative Removal by Adiabatic Gate: an RF pulse shaping technique using a quadrature derivative envelope to suppress state leakage during fast gates.
Josephson Junction	A quantum non-linear inductor formed by two superconducting electrodes separated by an ultra-thin insulating oxide tunnel barrier (Al-AlOx-Al).
JTWPA	Josephson Traveling Wave Parametric Amplifier: a wideband, quantum-noise-limited amplifier based on four-wave mixing along a nonlinear Josephson line.
OFHC Copper	Oxygen-Free High Thermal Conductivity copper (C10100): copper alloy with >99.99% purity utilized for cryogenic packaging due to high RRR and thermal conductance.
Purcell Effect	The enhanced spontaneous radiative decay of a qubit into the continuous impedance of an external microwave transmission line mediated by a resonator.
pyEPR	Python Energy Participation Ratio: an automated method to quantize non-linear quantum circuits directly from linear electromagnetic eigenmode solutions.
QICK	Quantum Instrumentation Control Kit: open-source FPGA firmware architecture (Xilinx RFSoC) performing direct digital microwave synthesis and readout.
SQUID	Superconducting Quantum Interference Device: a parallel loop of two Josephson junctions allowing magnetic flux tuning of the effective critical current.
Surface Code	A 2D topological quantum error-correcting code in which physical qubits are arranged on a square lattice with nearest-neighbor syndrome measurements.
TLS Loss	Two-Level System loss: microwave dielectric dissipation caused by quantum defects at material interfaces (substrate-metal, metal-air, substrate-air).
Through-Silicon Via (TSV)	A vertical metallized via hole etched through the silicon substrate to provide ultra-low-inductance microwave ground connections.

End of Technical Monograph